

SCHMIDT CIRCLES AND ELLIPTIC UNITS: Kronecker Limit Formulas and the Robert Index

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ABSTRACT. The phases u_f of the Gaussian Schmidt arrangement — the sixth coordinate on $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{C}) / \mathrm{SL}_2(\mathbb{Z})$, quantized on the Bianchi group in the companion paper — are, classwise, twisted Δ -quotients at CM points. We pursue this to its quantitative end. First, the character sums $S(\chi) = \sum \chi \log |u|$ over the classes of a level obey Kronecker limit formulas in both aspects of the theory: $S(\chi) = -2L'(0, \chi) + \frac{2}{3}\Sigma_0(\chi) + \frac{1}{2}\Sigma_{1728}(\chi)$ for the Euclidean phases of the curvature- $2n$ disks (over the ring class group of $\mathbb{Z} + n\mathbb{Z}[i]$), and, for the hyperbolic phases of discriminant $1 - n^2$, $S(\chi) = 0$ or $-4L'(0, \chi) + \frac{4}{3}\Sigma_0 + \Sigma_{1728}$ according to the value of χ on the twist class, the unit $n + \sqrt{n^2 - 1}$ cancelling identically. Here $L(s, \chi)$ is the Epstein L -function of the level and $\Sigma_x(\chi) = \sum \chi \log |j - x|$ a Gross–Zagier collision quantity. For genus characters everything is in closed form through Sturm-proved factorizations of $L(s, \chi)$; the real quadratic field carrying the Euclidean genus sum at conductor n is $\mathbb{Q}(\sqrt{n})$, and the j -dressings factor exactly in it with split-prime support at Gross–Zagier primes. Second, the $\mathfrak{r}$ -twisted Δ -ratios $R_f = r_0^6 \Delta(\mathfrak{b}_1) / \Delta(\mathfrak{r}^{-1} \mathfrak{b}_1)$ are algebraic units for every class of every odd level, imprimitive strata included — a local rigidity argument for isogenies attached to invertible ambiguous ideals, needing no quasi-canonical-lifting input — and the Euclidean Δ -data $n^{12} \Delta(\Lambda_c) / \Delta(\mathbb{Z}[i])$ have a $\mathfrak{P}$ -adic valuation independent of the class, with an explicit value that settles the split-prime denominator ladders. A canonical sixth root w_f obeys the laws at first power; at level 7 it is $-(2 \pm \sqrt{3})$. Third, on the cubic layer of the Euclidean theory the Δ -coset unit θ_u has index $[\mathcal{O}_{L_3}^\times : \langle -1, \theta_u \rangle] = 8 h_{L_3} C_n(0)$ in the unit group of the real cubic subfield L_3 of the ring class field, certified at all eight levels $n \leq 27$ with a cubic character, with $h_{L_3} = 1, 1, 3, 2$ at the primitive levels 9, 11, 13, 23, the last out of sample: the class number of the cubic field is the Robert index of the Schmidt units. Finally, the group of all ratios of the Δ -data has index $(4 \cdot 24^{h-1} / w) h_{H_n} \prod_\chi C_\chi(0)$ in the unit group of the ring class field H_n for every level — a Dedekind group determinant against the class number formula at $s = 0$ — verified with PARI/GP down to the exact exponent matrices, which show that modulo 24-th roots the index is exactly the class number times the imprimitivity multipliers; the hyperbolic units have the analogous index in the odd units of their class field, 24^6 exactly at level 21. Every finite claim is machine-verified; the scripts and precision policies are catalogued in an appendix.

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1. INTRODUCTION

Let $\mathcal{S} = \mathrm{PSL}_2(\mathbb{Z}[i]) \cdot \widehat{\mathbb{R}}$ be the Gaussian Schmidt arrangement, the orbit of the extended real line under the Bianchi group [21, 24, 25]. The companion paper [8] — cited throughout as [SchmidtI], with its theorem numbers — stratified $\mathcal{S}$ by an integer level, counted it in three geometries, computed the involution $\sigma(X) = \bar{X}^{-1}$ on the form classes of a level, and constructed a *phase*: a sixth coordinate on the double cosets $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{C}) / \mathrm{SL}_2(\mathbb{Z})$ which, on Schmidt matrices, quantizes to algebraic numbers u_f indexed by

the primitive form classes f of discriminant $1 - n^2$, and to Euclidean phases $u_{\mathfrak{c}}$ indexed by the ring class group $\text{Cl}(\mathbb{Z} + n\mathbb{Z}[i])$ of the curvature- $2n$ disks. Its last structural statement was the closed form

$$u_f = -\varepsilon \mu^{-2} \frac{h_2(\mathfrak{b}_1)}{h_2(\mathfrak{b}_2)}, \quad u_{\mathfrak{c}}^6 = -\beta^4(\beta - 1728)^3 \frac{\Delta(\Lambda_{\mathfrak{c}})}{\Delta(\mathbb{Z}[i])},$$

[8][Prop. 6.10, eq. (7.1)]: the phase is a weight-2 ratio at two CM points, and its sixth power a Δ -quotient dressed by singular moduli. This paper takes that statement to its quantitative end. The theme is single: *the phases of the Schmidt arrangement form an elliptic-unit system, and the index of that system is a class number.*

The arc. Write, for a class $\mathfrak{c}$ of a level with reduced CM point $\tau_{\mathfrak{c}}$, $g_{\mathfrak{c}} = y_{\mathfrak{c}}^6 |\Delta_q(\tau_{\mathfrak{c}})|$ for its scale-invariant Δ -datum, and, for a character χ of the class group,

$$S(\chi) = \sum_{\mathfrak{c}} \chi(\mathfrak{c}) \log |u_{\mathfrak{c}}|, \quad \Sigma_x(\chi) = \sum_{\mathfrak{c}} \chi(\mathfrak{c}) \log |j(\mathfrak{c}) - x|.$$

Taking logarithms of the closed forms and summing against χ , the first Kronecker limit formula converts $\sum \chi \log g$ into $-12L'(0, \chi)$ for the Epstein L -function of the level (Proposition 2.2), and one obtains the two *master identities* (Theorems 2.5 and 2.9):

$$\begin{aligned} \text{Euclidean: } S(\chi) &= -2L'(0, \chi) + \frac{2}{3}\Sigma_0(\chi) + \frac{1}{2}\Sigma_{1728}(\chi); \\ \text{hyperbolic: } S(\chi) &= \begin{cases} 0, & \chi(\mathfrak{r}_n) = +1, \\ -4L'(0, \chi) + \frac{4}{3}\Sigma_0(\chi) + \Sigma_{1728}(\chi), & \chi(\mathfrak{r}_n) = -1. \end{cases} \end{aligned}$$

The L' -part is the Kronecker limit formula; the j -parts are Gross–Zagier collision quantities. Two structural facts make the hyperbolic identity possible: the Norm Lemma of [8][Lemma 6.11] kills both the unit $\varepsilon = n + \sqrt{n^2 - 1}$ and the cocycle scalar μ in $|u_f|$, leaving a scale-invariant two-CM-point law (Lemma 2.8); and the twist law $u_f u_{\mathfrak{r}_n f} = 1$ of [8][Thm. 6.6] forces the character sums to live on the odd characters.

For genus characters the identities evaluate completely (Section 3). The Epstein L -function of a real character factors as a product of two Dirichlet L -functions times a finite Euler correction at square conductors — an identity of q -expansions, hence provable by Sturm’s bound — and $L'(0, \chi)$ becomes an explicit rational multiple of $\log \varepsilon_{d_2}$ for a fundamental unit. On the Euclidean side the real quadratic field is $\mathbb{Q}(\sqrt{n})$ in every case: the phases of the curvature- $2n$ disks know the Pell unit of $\mathbb{Q}(\sqrt{n})$. The j -dressings Σ_0, Σ_{1728} factor exactly in the same field as unit powers times ratios of split-prime generators, supported on the Gross–Zagier sets $\text{GZ}(D, -3)$ and $\text{GZ}(D, -4)$; for the non-real characters certified integer-relation searches find no such expression, as the Stark philosophy predicts.

The Δ -part of the hyperbolic phase is a genuine unit. With $r_0 = \frac{n-1}{2}$ and $\mathfrak{r} = [r_0, \sqrt{D}/2]$ the invertible ambiguous twist ideal of the order of discriminant $D = 1-n^2$, the $\mathfrak{r}$ -twisted Δ -ratio $R_f = u_f^6 \beta_2^4 (\beta_2 - 1728)^3 / \beta_1^4 (\beta_1 - 1728)^3$ has the lattice form $R_f = r_0^6 \Delta(\mathfrak{b}_1) / \Delta(\mathfrak{r}^{-1} \mathfrak{b}_1)$ (Lemma 4.1), and Theorem 4.2 proves that R_f is an algebraic unit for every class of every odd level, on every imprimitive stratum as well. The proof is local: the twist is an isogeny whose tangent valuation is the same for every proper ideal (a rigidity of p -divisible-group quotients), and composing two twists gives multiplication by p^k , which balances the valuation exactly against r_0^6 . No Kronecker limit formula, no Siegel units and no quasi-canonical liftings enter. The same rigidity gives the per-class valuation law for the Euclidean Δ -data (Theorem 4.11): $v_{\mathfrak{P}}(n^{12} \Delta(\Lambda_{\mathfrak{c}}) / \Delta(\mathbb{Z}[i]))$ is independent of $\mathfrak{c}$, with an explicit value $w_p(k)$, which settles the split-prime denominator ladders of [8][§7] exactly and decomposes the inert ones into certified single-slope ingredients. A canonical sixth root w_f of R_f obeys the mirror and twist laws at first power and is an η -quotient on $\Gamma_0(r_0)$ at a Heegner point (Section 5); at level 7 the first-power Schmidt units are $w = -(2 \pm \sqrt{3})$, the fundamental unit of $\mathbb{Q}(\sqrt{3})$ up to sign.

Finally, the index (Section 6). On the quadratic layer, the eigenprojections of the unit systems under a real character χ are explicit integer multiples of $\log \varepsilon_{d_2}$, with coefficients built from the class numbers $h(d_1)$ and $h(d_2)$. On the cubic layer — the eight Euclidean levels $n \leq 27$ whose ring class group has 3-torsion — the principal coset product θ of the Δ -data, normalized by the cube root of the Δ -mass, is a unit θ_u of the real cubic subfield L_3 of the ring class field, and

$$[\mathcal{O}_{L_3}^\times : \langle -1, \theta_u \rangle] = 8 h_{L_3} C_n(0)$$

(Theorem 6.6), where $C_n(0)$ is the integer imprimitivity multiplier, equal to 1 at the primitive levels 9, 11, 13, 23 where $h_{L_3} = 1, 1, 3, 2$. The identity $\text{index} = 8L'(0, \chi_3) / R_{L_3}$ is proved (Propositions 6.2–6.4); the content of the theorem is the certified computation of the fundamental units, rigorous through a root descent capped by Friedman’s regulator bound. The row $n = 23$ was computed after the law $8h_{L_3}$ had been formulated on 9, 11, 13, and confirmed it on a new cubic field of class number 2: *the class number of the cubic field is the Robert index of the Schmidt units*, the exact analogue of “cyclotomic units have index h^+ ”, realized by the disks of curvature $2n$.

The cubic layer is one eigen-piece of a single object, and the present version of the paper closes the loop: the group $\mathcal{V}_n$ of all ratios $G_{\mathfrak{c}}/G_{\mathfrak{c}'}$ of the Δ -data is a subgroup of full rank of the unit group of the ring class field H_n , and its regulator is a Dedekind group determinant, so that (Theorem 6.7)

$$[\mathcal{O}_{H_n}^\times : \mu(H_n) \mathcal{V}_n] = \frac{4 \cdot 24^{h-1}}{w_{H_n}} h_{H_n} \prod_{\chi \neq 1} C_\chi(0)$$

for *every* level $n \geq 2$: the class number of the ring class field, times an explicit power of 2 and 3, times the Euler multipliers of the imprimitive characters. The same argument gives the index in every subfield cut out by a subgroup of the class group (Theorem 6.12). The theorem is verified with PARI/GP at all levels $n \leq 15$ (unconditionally) and at $n = 23$, down to the exact exponent matrices of the units in a fundamental-unit basis, which reveal that modulo 24-th roots the index is *exactly* $h_{H_n} \prod C_\chi(0)$ (Remark 6.15, Conjecture 6.16).

Position in the literature. The objects are classical and the ambition is to place a new geometric invariant among them. Kronecker’s limit formula and its use to produce units in class fields of imaginary quadratic fields go back to Kronecker and Hecke, in the form we use to Siegel’s lectures [23] and Ramachandra [18]; Robert [19] defined the elliptic units of an abelian extension of an imaginary quadratic field and proved that their index in the full unit group is the class number up to an explicit factor, refined by Gillard–Robert [4] and systematized, with the function-theoretic side of Siegel and Klein forms, by Kubert–Lang [10, Ch. 11–13]; index formulas for ramified elliptic units in the style of Sinnott are due to Oukhaba [17]; Schertz’s monograph [20, Ch. 6–7] presents the relation between elliptic units and class numbers and the Stark units of ring class fields. Stark [26] showed that his conjecture over an imaginary quadratic base is a theorem through elliptic units; for ring class fields of prime conductor the Stark unit generates, with the roots of unity, a subgroup of index exactly the class number of the ring class field — the form in which Hajir and Rodriguez Villegas [7] made the units explicit and in which Küçüksakallı [11] computed class numbers of ring class fields. On the finite-place side, Gross–Zagier’s factorization of singular moduli [6] is the reference for the collision quantities Σ_0 , Σ_{1728} , and Gross’s quasi-canonical liftings [5] give the conceptual reading of the valuation laws of Section 4 — a reading we keep as interpretation, not input. Meyer’s monograph [15] computed class numbers of abelian extensions of quadratic fields through Kronecker limit formulas, and the higher Kronecker limit formulas of Vlasenko–Zagier [30] along real quadratic data are the closest existing relatives of $\log |u_f|$ on the real-quadratic side, where the phase is normalized by the unit of $\mathbb{Q}(\sqrt{n^2 - 1})$ and was shown in [8][Prop. 6.29] not to be cycle-integral data. Friedman’s regulator bound [3] is the tool that makes the fundamental units of Section 6 rigorous.

What is new is not the Kronecker limit formula, nor the existence of units in ring class fields with class-number index — those are the classical theory just cited. It is that a *geometric* invariant of circles is such a system: the character sums of the phase compute $L'(0, \chi)$ exactly, with the unit ε cancelling and the genus field $\mathbb{Q}(\sqrt{n})$ appearing; the ambiguous-ideal twist $\mathfrak{r}$ of a non-maximal order preserves every local Δ -datum, so that the Δ -ratios the phase carries are units on every stratum; the Euclidean Δ -data have class-independent valuations; and the Δ -coset units of the curvature- $2n$

disks have index $8h_{L_3}C_n(0)$ in the cubic layer over $\mathbb{Q}(i)$, a number computed and certified at eight levels. A literature record accompanies the paper in the repository; we found no prior index computation for elliptic-unit systems in ring class fields of $\mathbb{Q}(i)$ and no prior appearance of these Δ -coset units. How θ_u compares with the Stark unit of the ring class field — both generate finite-index subgroups, with indices $8h_{L_3}$ in $\mathcal{O}_{L_3}^\times$ and h_{H_n} in $\mathcal{O}_{H_n}^\times$ respectively — is a natural question we do not pursue here.

What is proved and what is certified. Proved for every level: the limit-value formula (Proposition 2.2); the classwise laws (Lemmas 2.4, 2.8); both master identities (Theorems 2.5, 2.9) with the trivial-character anchor; the Δ -mass polynomial and the uniform Stark law (Theorem 2.6); the rationality of the twisted-ratio polynomial (Theorem 2.10); the genus factorizations and the resulting $L'(0, \chi_2)$ closed forms at every displayed level (Proposition 3.1, Sturm’s bound); each displayed Σ -factorization (exact arithmetic in its quadratic field); the lattice lemma and the unit theorem on all strata (Lemma 4.1, Theorem 4.2); the collapse $R \equiv 1$ on strata with principal induced twist; the per-class valuation law and the split ladder (Theorem 4.11, Corollary 4.13); the first-power laws of w_f and its η -quotient form (Proposition 5.2); the unit property of θ_u , the Stark relation and the index identity $8L'(0, \chi_3)/R_{L_3}$ (Propositions 6.2–6.4); the full Robert index of the Δ -units and its layer form (Theorems 6.7, 6.12, with Lemmas 6.8–6.11). Certified, by exact integer arithmetic or with at least 140 (typically ≥ 200) spare digits, and not proved in general: the coefficient values of the displayed polynomials and their irreducibility; the Newton-polygon verification of Theorem 4.11 and the inert-ladder decompositions; the coherence table $m(n)$ of the first-power units; the fundamental units, class numbers and indices of Theorem 6.6; the PARI record of Proposition 6.14 (unconditional at $n \leq 15$, GRH at $n = 23$) and the saturation identities of Remark 6.15; the ε -ratio identifications at the composite conductor $n = 15$; and the converse direction of the stratum criterion. Certified negatives: the PSLQ non-fits for non-real characters and the absence of a congruence law for $m(n)$.

Plan of the paper. Section 2 proves the limit-value formula, the classwise laws and the master identities, and records the Δ -mass and twisted-ratio polynomials. Section 3 evaluates the genus characters in closed form. Section 4 proves the unit theorem and the per-class valuation law — the technical core. Section 5 constructs the first-power Schmidt units. Section 6 computes the Robert index, on the cubic layer and in full. Section 7 lists the four genuine continuations, and Appendix A documents the machine verification.

Status conventions. As in [8], a **Theorem**, **Proposition**, **Lemma** or **Corollary** without qualification is proved. A statement labelled (*computational*) is a finite claim verified by exact arithmetic, or a numerical claim certified at a stated precision with stated safety margins, by a named script;

the range of verification is part of the statement. Every script cited was re-run for this paper (Appendix A). No integer-relation search is used to establish a claim: the discoveries made by PSLQ are re-derived in exact arithmetic, and PSLQ appears only in certified *negative* statements.

Notation. K denotes the imaginary quadratic field of the aspect at hand. *Euclidean level* $n \geq 2$: $K = \mathbb{Q}(i)$, $\mathcal{O}_n = \mathbb{Z} + n\mathbb{Z}[i]$ the order of conductor n , discriminant $D = -4n^2$, class group $\text{Cl}(\mathcal{O}_n)$ of order $h = h(-4n^2)$, ring class field H_n ; the phases $u_\mathfrak{c} = \Theta/\Omega$ of [8][§7], normalized by the lemniscatic period, with $|u_\mathfrak{c}|$ and $u_\mathfrak{c}^2$ well defined on classes [8][Prop. 7.10]; $\Lambda_\mathfrak{c} \subseteq \mathbb{Z}[i]$ the primitive index- n sublattice of the class [8][Thm. 7.6]; $G_\mathfrak{c} := n^{12}\Delta(\Lambda_\mathfrak{c})/\Delta(\mathbb{Z}[i])$ and the Δ -mass $M(n) = \prod_\mathfrak{c} G_\mathfrak{c}$ [8][Thm. 7.18]. *Hyperbolic level* $n \geq 3$ odd: $K = \mathbb{Q}(\sqrt{1-n^2})$, $D = 1-n^2$, $N = n^2-1$, $\mathcal{O}$ the order of discriminant D with class group $\text{Cl}(D)$ of primitive classes, $h = h(D)$, ring class field H ; $\varepsilon = n + \sqrt{N}$; $r_0 = \frac{n-1}{2}$, $s_0 = \frac{n+1}{2}$, $\omega_0 = \sqrt{D}/2$; the twist ideal $\mathfrak{r} = \mathfrak{r}_n = [r_0, \omega_0]$, invertible and ambiguous, with $\bar{\mathfrak{r}} = \mathfrak{r}$, $\mathfrak{r}^2 = (r_0)$, $N\mathfrak{r} = r_0$, and class $[\mathfrak{r}_n] = [(r_0, 0, s_0)]$ [8][Thm. 4.5]; for a primitive class f with CM point m_1 , $\mathfrak{b}_1 = [1, m_1]$, $\beta_1 = j(f)$, $\beta_2 = j(\mathfrak{r}f) = j(\mathfrak{r}^{-1}\mathfrak{b}_1)$ (the complex conjugate of the moduli coordinate β_2 of [8][§6.1]), and $u_f = \varepsilon\Theta_f$ the normalized phase with laws $u_{f^{-1}} = \bar{u}_f$ and $u_f u_{\mathfrak{r}f} = 1$ [8][Thm. 6.6]. In both aspects the unit count of the order is $w = 2$. $\Delta_q = \eta^{24}$ is the normalized discriminant, $\Delta(\Lambda)$ the weight-12 lattice discriminant, j the modular invariant, $\tau_\mathfrak{c}$ the CM point of the reduced form $Q_\mathfrak{c} = (A, B, C)$ of a class, $y_\mathfrak{c} = \text{Im } \tau_\mathfrak{c} = \sqrt{|D|}/2A$, and

$$(1.1) \quad g_\mathfrak{c} := y_\mathfrak{c}^6 |\Delta_q(\tau_\mathfrak{c})|$$

the scale-invariant Δ -datum of the class ($g = \text{covol}^6 |\Delta|$ of any lattice in the class, up to one global constant). For a real quadratic discriminant d , $\varepsilon_d > 1$ is the fundamental unit and $h(d)$ the class number; $w(d)$ is the number of roots of unity of the imaginary quadratic order of discriminant $d < 0$.

Acknowledgements. The results of this paper were obtained in a sequence of computer-assisted research sessions; the complete research notes, verification scripts and literature record are available in the accompanying repository.

2. THE CLASSWISE LAWS AND THE MASTER IDENTITIES

2.1. The Epstein L -function of a level and the limit value.

Definition 2.1. For a discriminant $D < 0$ with class group $\text{Cl}(D)$ of primitive forms and a character χ of $\text{Cl}(D)$, set

$$L(s, \chi) := \frac{1}{w} \sum_{\mathfrak{c} \in \text{Cl}(D)} \chi(\mathfrak{c}) \zeta_{Q_\mathfrak{c}}(s), \quad \zeta_Q(s) = \sum'_{(x,y) \in \mathbb{Z}^2} Q(x,y)^{-s},$$

the form-class (Epstein) L -function of the discriminant. For $m \geq 1$ let $r_{\mathfrak{c}}(m)$ be the number of representations of m by $Q_{\mathfrak{c}}$ and $R_{\chi}(m) := \sum_{\mathfrak{c}} \chi(\mathfrak{c}) r_{\mathfrak{c}}(m)$, so that $L(s, \chi) = w^{-1} \sum_{m \geq 1} R_{\chi}(m) m^{-s}$.

Since $(x, y) \mapsto (x, -y)$ bijects the representations by Q and by Q^{-1} , one has $R_{\chi} = R_{\bar{\chi}}$, and $L(s, \chi) = L(s, \bar{\chi})$ is real on the real axis. In our two aspects $D = -4n^2$ resp. $D = 1 - n^2$, and $w = 2$.

Proposition 2.2 (functional equation and the limit value). *Write $\gamma(s) = (\frac{\sqrt{|D|}}{2\pi})^s \Gamma(s)$. Then $\gamma(s) \zeta_Q(s) = \gamma(1-s) \zeta_Q(1-s)$ for every class, so for nontrivial χ the function $L(s, \chi)$ is entire with $L(0, \chi) = 0$, and*

$$(2.1) \quad L'(0, \chi) = \frac{\sqrt{|D|}}{2\pi} L(1, \chi) = -\frac{1}{12} \sum_{\mathfrak{c}} \chi(\mathfrak{c}) \log g_{\mathfrak{c}}.$$

Proof. With $\tau_Q = \frac{-B+i\sqrt{|D|}}{2A}$ one has $A|x + y\tau_Q|^2 = Q(x, -y)$ and $Ay\tau_Q = \sqrt{|D|}/2$, whence $\zeta_Q(s) = (\frac{2}{\sqrt{|D|}})^s E(\tau_Q, s)$ for the nonholomorphic Eisenstein series $E(\tau, s) = \sum' y^s / |m\tau + n|^{2s}$. The completed series $E^*(\tau, s) = \pi^{-s} \Gamma(s) E(\tau, s)$ satisfies $E^*(\tau, s) = E^*(\tau, 1-s)$, and $\gamma(s) \zeta_Q(s) = E^*(\tau_Q, s)$: this is the functional equation, class by class. For nontrivial χ the poles at $s = 1$ cancel in the character sum, so $\Gamma(s) L(s, \chi)$ is finite at $s = 0$; since $\Gamma(s) \sim 1/s$, $L(0, \chi) = 0$ and $L'(0, \chi) = \lim_{s \rightarrow 0} \gamma(s) L(s, \chi) = \gamma(1) L(1, \chi)$. The first Kronecker limit formula [23, Ch. I], [12, Ch. 20],

$$E(\tau, s) = \frac{\pi}{s-1} + 2\pi(\gamma_E - \log 2 - \log(\sqrt{y} |\eta(\tau)|^2)) + O(s-1),$$

gives, after the χ -sum has killed every $\mathfrak{c}$ -independent constant, and with $\sqrt{y} |\eta|^2 = (y^6 |\Delta_q|)^{1/12} = g^{1/12}$,

$$L(1, \chi) = -\frac{1}{w} \cdot \frac{2}{\sqrt{|D|}} \cdot \frac{\pi}{6} \sum_{\mathfrak{c}} \chi(\mathfrak{c}) \log g_{\mathfrak{c}};$$

multiply by $\gamma(1) = \sqrt{|D|}/2\pi$ and use $w = 2$. $\square$

The verification of every identity below rests on an evaluation of $L'(0, \chi)$ that uses *no* modular quantity at all.

Lemma 2.3 (independent evaluation). *For nontrivial χ , with $\alpha = 2\pi/\sqrt{|D|}$ and E_1 the exponential integral,*

$$L'(0, \chi) = \frac{1}{w} \sum_{m \geq 1} R_{\chi}(m) \left[\frac{e^{-\alpha m}}{\alpha m} + E_1(\alpha m) \right].$$

Proof. Let $\theta_Q(t) = \sum_{(x,y) \in \mathbb{Z}^2} e^{-\alpha Q(x,y)t}$. The Mellin transform gives

$$\gamma(s) \zeta_Q(s) = \int_0^\infty (\theta_Q(t) - 1) t^{s-1} dt,$$

and Poisson summation gives $\theta_Q(1/t) = t\theta_{Q^{-1}}(t)$: in the normalization $\alpha = 2\pi/\sqrt{|D|}$ the Gram matrix $2G_Q/\sqrt{|D|}$ has determinant 1, and its inverse is $2G_{Q^{-1}}/\sqrt{|D|}$. Splitting the integral at $t = 1$ and inverting t on $(0, 1)$,

$$\gamma(s)\zeta_Q(s) = \int_1^\infty (\theta_Q(t) - 1)t^{s-1} dt + \int_1^\infty (\theta_{Q^{-1}}(t) - 1)t^{-s} dt + \frac{1}{s-1} - \frac{1}{s}.$$

Summing against a nontrivial χ kills the last two terms, and $R_\chi = R_{\bar{\chi}}$ identifies the Q^{-1} -sum with the Q -sum; at $s = 0$ the left side is $L'(0, \chi)$ by Proposition 2.2, and the two integrals of $e^{-\alpha mt}$ are $E_1(\alpha m)$ and $e^{-\alpha m}/\alpha m$. $\square$

The series converges like $e^{-\alpha m}$; at 250 digits it needs between 600 and 2600 terms at the levels below, and its inputs are the exact integers $R_\chi(m)$. Every limit-formula identity in this paper is checked against it (residuals $\leq 2 \cdot 10^{-248}$ at 250 digits; Appendix A).

2.2. Euclidean: the classwise law and the master identity.

Lemma 2.4 (classwise law). *For every class $\mathfrak{c} \in \text{Cl}(\mathcal{O}_n)$ of every level $n \geq 2$, with $\beta = j(\mathfrak{c})$,*

$$|u_{\mathfrak{c}}|^6 = |\beta^4(\beta - 1728)^3| \frac{g_{\mathfrak{c}}}{n^6 |\Delta_q(i)|}.$$

Proof. $u^6 = -\beta^4(\beta - 1728)^3 \Delta(\Lambda_{\mathfrak{c}})/\Delta(\mathbb{Z}[i])$ is [8][eq. (7.1)]. Write $\Lambda_{\mathfrak{c}} = c[1, w]$; then $|c|^2 y_w = n = \text{covol}(\Lambda_{\mathfrak{c}})$ and $|\Delta(\Lambda_{\mathfrak{c}})| = |c|^{-12} (2\pi)^{12} |\Delta_q(w)|$, so $|\Delta(\Lambda_{\mathfrak{c}})/\Delta(\mathbb{Z}[i])| = y_w^6 |\Delta_q(w)|/(n^6 |\Delta_q(i)|)$, and $y_w^6 |\Delta_q(w)| = g_{\mathfrak{c}}$ by $\text{SL}_2(\mathbb{Z})$ -invariance of $y^6 |\Delta_q|$. $\square$

Theorem 2.5 (Euclidean master identity). *For every nontrivial character χ of $\text{Cl}(\mathcal{O}_n)$, with $S(\chi) = \sum_{\mathfrak{c}} \chi(\mathfrak{c}) \log |u_{\mathfrak{c}}|$ and $\Sigma_x(\chi) = \sum_{\mathfrak{c}} \chi(\mathfrak{c}) \log |j(\mathfrak{c}) - x|$,*

$$(2.2) \quad \boxed{S(\chi) = -2L'(0, \chi) + \frac{2}{3}\Sigma_0(\chi) + \frac{1}{2}\Sigma_{1728}(\chi).}$$

For the trivial character, with H_{-4n^2} the ring class polynomial and $M(n)$ the Δ -mass of [8][Thm. 7.18],

$$(2.3) \quad 6 \sum_{\mathfrak{c}} \log |u_{\mathfrak{c}}| = 4 \log |H_{-4n^2}(0)| + 3 \log |H_{-4n^2}(1728)| + \log |M(n)| - 12h \log n.$$

Proof. Take logarithms in Lemma 2.4 and sum against χ : the $\mathfrak{c}$ -independent constant $-6 \log n - \log |\Delta_q(i)|$ dies for $\chi \neq 1$, and $\sum \chi \log g = -12L'(0, \chi)$ by Proposition 2.2. For the trivial character sum the same identity over all classes: $\sum \log |\beta| = \log |H(0)|$, $\sum \log |\beta - 1728| = \log |H(1728)|$, and $\sum_{\mathfrak{c}} (\log g_{\mathfrak{c}} - \log |\Delta_q(i)| - 6 \log n) = \log |M(n)| - 12h \log n$ by the definition $M(n) = \prod_{\mathfrak{c}} n^{12} \Delta(\Lambda_{\mathfrak{c}})/\Delta(\mathbb{Z}[i])$. $\square$

Both statements are verified at $n = 3, 5, 7, 9, 11, 13, 15$, every character, with residuals $\leq 2 \cdot 10^{-248}$ (250 digits; 400 at $n = 15$), the identity and

the independent evaluation of Lemma 2.3 being separate computations; the trivial-character anchor is checked against the certified H_{-4n^2} and the closed form of $M(n)$.

The limit formula becomes an algebraic statement through one integer polynomial per level.

Theorem 2.6 (the Δ -mass polynomial and the uniform Stark law). *Let $G_{\mathfrak{c}} = n^{12}\Delta(\Lambda_{\mathfrak{c}})/\Delta(\mathbb{Z}[i])$; this is a nonzero algebraic integer, of absolute value $n^6 g_{\mathfrak{c}}/|\Delta_q(i)|$. Then:*

- (1) $\sigma(G_{\mathfrak{c}}) = G_{\mathfrak{c}(\sigma)^{-1}\mathfrak{c}}$ for $\sigma \in \text{Gal}(\bar{\mathbb{Q}}/K)$ with Artin class $\mathfrak{c}(\sigma)$ on H_n , and $\overline{G_{\mathfrak{c}}} = G_{\mathfrak{c}^{-1}}$; hence the multiset $\{G_{\mathfrak{c}}\}$ is $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ -stable, and

$$D_n(x) := \prod_{\mathfrak{c} \in \text{Cl}(\mathcal{O}_n)} (x - G_{\mathfrak{c}}) \in \mathbb{Z}[x], \quad D_n(0) = (-1)^h M(n).$$

- (2) For every nontrivial χ : $-12 L'(0, \chi) = \sum_{\mathfrak{c}} \chi(\mathfrak{c}) \log |G_{\mathfrak{c}}|$.

Thus all the $L'(0, \chi)$ of a level are character combinations of the logarithms of the roots of one integer polynomial: the Δ -data $\{G_{\mathfrak{c}}\}$ of the curvature- $2n$ disks form an elliptic-unit system for the ring class tower of $\mathbb{Q}(i)$ in the sense of Siegel and Robert.

Proof. (1) $G_{\mathfrak{c}} = F(\Lambda_{\mathfrak{c}}; \mathbb{Z}[i])$ with $F(\Lambda; L) = n^{12}\Delta(\Lambda)/\Delta(L)$, isobaric of joint weight zero. The transport lemma [8][Lemma 7.13] gives $\sigma(F(\Lambda_{\mathfrak{c}}; \mathbb{Z}[i])) = F(s^{-1}\Lambda_{\mathfrak{c}}; s^{-1}\mathbb{Z}[i])$ for the idèle s of σ ; rescaling jointly by a generator of the principal ideal $(s^{-1}\mathbb{Z}[i])^{-1}$, exactly as in the translation law [8][Prop. 7.14], lands on a primitive representative of the class $\mathfrak{c}(\sigma)^{-1}\mathfrak{c}$, determined up to μ_4 — and $\Delta(i\Lambda) = \Delta(\Lambda)$, so there is no ambiguity at all. Complex conjugation: $\Delta(\bar{\Lambda}) = \overline{\Delta(\Lambda)}$ and $\bar{\Lambda}$ represents $\mathfrak{c}^{-1}$ [8][Prop. 7.10]. The coefficients of D_n are therefore rational, and integral because $n^{12}\Delta$ -quotients of commensurable CM lattices are algebraic integers [12, Ch. 12]; $D_n(0) = (-1)^h \prod G_{\mathfrak{c}} = (-1)^h M(n)$. (2) $\log |G_{\mathfrak{c}}| = \log g_{\mathfrak{c}} + 6 \log n - \log |\Delta_q(i)|$, so the character sum reproduces Proposition 2.2. $\square$

The certified polynomials (integer coefficients with ≥ 226 spare digits at 250 digits, 358 at $n = 15$; constant terms equal to $(-1)^h M(n)$ exactly):

$$\begin{aligned} D_3 &= x^2 - 378x + 729, & D_5 &= x^2 - 322x + 1, \\ D_7 &= x^4 + 58604x^3 + 5618502582x^2 - 191605917748x + 7^6, \\ D_9 &= x^6 + 777114x^5 + 469570343535x^4 + 88033943990172204x^3 \\ &\quad + 4975970841324612804303x^2 - 6464018128844293554150x + 3^{24}, \\ D_{11} &= x^8 - 2940366x^5 + 2977596073815x^4 + 116499046538184860x^3 \\ &\quad + 38813935158542440699935x^2 - 1046289507684469650030x + 11^6, \\ D_{13} &= x^6 + 7210938x^5 + 12700046391567x^4 - 1529217345755339156x^3 \\ &\quad + 61048319821249936271631x^2 - 22813565655771828294x + 1, \end{aligned}$$

$$\begin{aligned}
D_{15} = & x^8 - 243432x^7 - 49497426055620x^6 - 1528089585117383803032x^5 \\
& + 12625658759774046395283174x^4 \\
& + 2831276695986378100679863409676072x^3 \\
& + 85367425855325247262124045173735929698940x^2 \\
& - 331767481864968679465735138096222248x + 3^{24}.
\end{aligned}$$

Remark 2.7 (coset cubics; the Stark objects of the cubic layer). At $n = 9, 11, 13$, where $\text{Cl}(\mathcal{O}_n) \cong \mathbb{Z}/6$, the products of the $G_{\mathfrak{c}}$ over the three cosets of the kernel of the order-3 character are the roots of the certified integer cubics

$$\begin{aligned}
n = 9 : & \quad x^3 - 76822296291x^2 + 4976085199686658256835x - 3^{24}, \\
n = 11 : & \quad x^3 + 277946532501x^2 + 38813938298955894507411x - 11^6, \\
n = 13 : & \quad x^3 + 446643445245x^2 + 61048319249786206560771x - 1,
\end{aligned}$$

with the Stark-type closed form $-12L'(0, \chi_3) = \frac{3}{2} \log |G_0^{(3)}| - \frac{1}{2} \log |C_{\Delta}(0)|$, $G_0^{(3)}$ the principal coset product and $C_{\Delta}(0)$ the constant term, verified to 10^{-247} . These cubics are the objects whose unit normalization carries the Robert index of Section 6.

2.3. Hyperbolic: cancellation of ε and μ , and the odd-character law.

Lemma 2.8 (scale-invariant closed form). *For every primitive class f of discriminant $D = 1 - n^2$, with $\beta_1 = j(f)$, $\beta_2 = j(\mathfrak{r}_n f)$ and $g(f) = y_f^6 |\Delta_q(\tau_f)|$,*

$$|u_f|^6 = \frac{|\beta_1^4(\beta_1 - 1728)^3|}{|\beta_2^4(\beta_2 - 1728)^3|} \cdot \frac{g(f)}{g(\mathfrak{r}_n f)}.$$

In particular neither $\varepsilon = n + \sqrt{n^2 - 1}$ nor μ survives in $|u_f|$.

Proof. By the closed form [8][Prop. 6.10], $u_f = -\varepsilon \mu^{-2} h_2(\mathfrak{b}_1)/h_2(\mathfrak{b}_2)$ with $\mathfrak{b}_2 = [1, -m_2]$ the lattice of the class $[\mathfrak{r}_n f]$ [8][Prop. 6.7], and by the Norm Lemma [8][Lemma 6.11], $|\mu|^2 = \varepsilon q_1/q_2$ with q_1, q_2 the half-curvatures of the two circles; hence $|u_f| = (q_2/q_1) |h_2(\mathfrak{b}_1)/h_2(\mathfrak{b}_2)|$. The weight algebra $h_2^6 = c j^4(j - 1728)^3 \Delta$ (one universal constant c) and

$$|\Delta(\mathfrak{b}_i)| = (2\pi)^{12} \frac{g(\mathfrak{b}_i)}{\text{covol}(\mathfrak{b}_i)^6}, \quad \frac{\text{covol}(\mathfrak{b}_1)}{\text{covol}(\mathfrak{b}_2)} = \frac{q_2}{q_1}$$

(both CM points have imaginary part $\sqrt{N}/2q_i$) give

$$\left| \frac{h_2(\mathfrak{b}_1)}{h_2(\mathfrak{b}_2)} \right|^6 = \left(\frac{q_1}{q_2} \right)^6 \cdot \frac{|\beta_1^4(\beta_1 - 1728)^3|}{|\beta_2^4(\beta_2 - 1728)^3|} \cdot \frac{g_1}{g_2};$$

assemble, and identify β_2, g_2 as the class functions at $\mathfrak{r}_n f$ through $[\mathfrak{b}_2] = [\mathfrak{r}_n][\mathfrak{b}_1]$ [8][Thm. 4.5]. $\square$

The lemma is verified classwise at $n = 9, 11, 13, 15$ with maximal deviation $4.5 \cdot 10^{-249}$. The expected new ingredient of the hyperbolic aspect, $\log \varepsilon_{n^2-1}$ coming from $\varepsilon^6 \mu^{-12}$, cancels identically: the Norm Lemma places ε exactly as the discrepancy between the two archimedean places of μ , and $|u_f|$ sees only one of them.

Theorem 2.9 (hyperbolic master identity). *Let χ be a character of $\text{Cl}(1 - n^2)$, $S(\chi) = \sum_f \chi(f) \log |u_f|$, $\Sigma_x(\chi) = \sum_f \chi(f) \log |\beta_1(f) - x|$. Then*

$$(2.4) \quad S(\chi) = \begin{cases} 0, & \chi(\mathfrak{r}_n) = +1, \\ -4L'(0, \chi) + \frac{4}{3}\Sigma_0(\chi) + \Sigma_{1728}(\chi), & \chi(\mathfrak{r}_n) = -1. \end{cases}$$

Proof. Sum $6 \log |u_f|$ from Lemma 2.8 against χ . Re-indexing the $\mathfrak{r}_n f$ -terms by $f \mapsto \mathfrak{r}_n f$ multiplies their sum by $\chi(\mathfrak{r}_n)^{-1} = \chi(\mathfrak{r}_n)$, the twist class being 2-torsion, so $6S(\chi) = (1 - \chi(\mathfrak{r}_n))[4\Sigma_0 + 3\Sigma_{1728} + \Sigma_\gamma]$ with $\Sigma_\gamma(\chi) = \sum_f \chi(f) \log g(f) = -12L'(0, \chi)$ by Proposition 2.2 at discriminant $1 - n^2$. $\square$

The vanishing statement is the twist law $u_{\mathfrak{r}_n f} u_f = 1$ seen through the limit formula; both branches are verified at $n = 9, 11, 13, 15$, every character, with residuals $\leq 10^{-248}$, the even-character sums vanishing to 10^{-249} or exactly.

Theorem 2.10 (the twisted Δ -ratio polynomial). *Set*

$$(2.5) \quad R_f := u_f^6 \frac{\beta_2^4(\beta_2 - 1728)^3}{\beta_1^4(\beta_1 - 1728)^3}, \quad |R_f| = \frac{g(f)}{g(\mathfrak{r}_n f)},$$

a class function, with $R_{\mathfrak{r}_n f} = R_f^{-1}$. Then the multiset $\{R_f\}_{f \in \text{Cl}(D)}$ is stable under $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$, so that $\prod_f (x - R_f) \in \mathbb{Q}[x]$; and for every odd character ($\chi(\mathfrak{r}_n) = -1$),

$$(2.6) \quad -24L'(0, \chi) = \sum_f \chi(f) \log |R_f|.$$

Proof. $|R_f| = g(f)/g(\mathfrak{r}_n f)$ is Lemma 2.8, and $R_{\mathfrak{r}_n f} = 1/R_f$ is the twist law together with $\mathfrak{r}^2 = 1$ in $\text{Cl}(D)$. By the first-power dihedral law [8][Thm. 6.19] and the reciprocity for singular moduli, u_f , β_1 and β_2 are transported by the same generalized dihedral permutation $f \mapsto f^{e(\sigma)} \mathfrak{c}(\sigma)$ of the classes (the translated pair is again the canonical pair of the translated class), so $\sigma(R_f) = R_{f^{e(\sigma)} \mathfrak{c}(\sigma)}$. Finally $\sum_f \chi(f) [\log g(f) - \log g(\mathfrak{r}_n f)] = (1 - \chi(\mathfrak{r}_n)) \Sigma_\gamma(\chi) = 2\Sigma_\gamma(\chi) = -24L'(0, \chi)$ for odd χ . $\square$

This is the hyperbolic counterpart of Theorem 2.6, and sharper: the certified polynomials $\prod_f (x - R_f)$ (Section 4.3) have *integer* coefficients, are palindromic, and have constant term 1 — the R_f are algebraic units, inverted in pairs by the $\mathfrak{r}_n$ -twist, so the $L'(0, \chi)$ of the odd characters are χ -combinations of logarithms of genuine units, with no S -integer dressing at all in the Δ -part. Theorem 4.2 proves this for every n .

2.4. The certified record. Character sums and L' -values at the computed levels (20 digits shown; computed at 250 digits, 400 at Euclidean $n = 15$; $S(\chi) = S(\bar{\chi})$, one value per conjugate pair; characters of a non-cyclic class group are indexed by their exponents on the generators of $\mathbb{Z}/4 \times \mathbb{Z}/2$). Euclidean:

n	$\text{ord } \chi$	$S(\chi)$	$L'(0, \chi)$
3	2	10.119185642834999673	0.43898596564160556954
5	2	16.401110452980199012	0.96242365011920689500
7	2	50.898733868194595161	2.7686593833135738327
7	4	22.684295755688513572	1.4860221248769270925
9	2	18.137584684821439049	1.7559438625664222782
9	3	70.144229691414912554	4.0476417264498781634
9	6	34.382429251891407159	2.1364594185869239177
11	2	50.213109260828194576	2.9932228461263808979
11	3	86.033492426467556694	5.3026856597888262543
11	6	27.769444924657409111	2.3032180902028192162
13	2	23.802362926332249068	2.3895264345742186082
13	3	101.24102291036467905	6.5582440777920829100
13	6	50.399596052674761580	3.3904644593763457318
15	2 (0, 1)	62.742974077314449734	3.8496946004768275800
15	2 (2, 0)	123.17609219442540179	8.2537482755822421869
15	2 (2, 1)	48.769335467586396360	3.5118877251328445563
15	4 (1, 0)	132.06845592069919001	8.2852528669601990706
15	4 (1, 1)	39.877919196363264725	3.4800432906097017186

Hyperbolic (odd characters; the even ones have $S = 0$ to 10^{-249} or exactly):

n	χ	$S(\chi)$	$L'(0, \chi)$
9	order 4	45.337283759443440194	1.0612750619050356520
11	(0, 1)	71.156367508225883636	1.7627471740390860505
11	(1, 0)	46.591472482040984242	1.2122976394880445490
13	(0, 1)	86.596498648832811158	2.2924316695611776878
13	(1, 0)	57.828173780263506754	1.5667992369724110787
15	(0, 1)	96.655012640029353845	2.7686593833135738327
15	(2, 1)	42.411391969797829839	1.7000422070566697504
15	order 4	101.30861654091162593	2.5318972768037898006

Two coincidences are visible to the eye and explained in Section 3: Euclidean $n = 7$ (order 2) and hyperbolic $n = 15$, $\chi_{(0,1)}$, share $L' = \log(8 + 3\sqrt{7})$, the Kronecker-symbol pairs $(-7) \cdot 28$ and $(-8) \cdot 28$ arriving at the same ε_{28} ; and the three real characters at Euclidean $n = 15$ have $L' = 4 \log \varepsilon_{60}$, $8 \log \varepsilon_5$ and $\frac{8}{3} \log \varepsilon_{12}$.

3. GENUS CHARACTERS IN CLOSED FORM

For a real (genus) character every term of the master identities evaluates in closed form: the L' -part through a factorization of the Epstein L -function

into two Dirichlet L -functions, the j -parts through exact factorizations in a real quadratic field.

3.1. Euclidean: the L' -part and the field $\mathbb{Q}(\sqrt{n})$. $\text{Cl}(\mathcal{O}_n)$ is cyclic of order 6 at $n = 9, 11, 13$, of order 2 at $n = 3, 5$, $\mathbb{Z}/4$ at $n = 7$, and $\mathbb{Z}/4 \times \mathbb{Z}/2$ at $n = 15$; so there is one nontrivial real character χ_2 per level $n \leq 13$ and three at $n = 15$. For a discriminant d write $\chi_d = (\frac{d}{\cdot})$ for the Kronecker symbol, a (possibly imprimitive) Dirichlet character, $L(s, \chi_d)$ for its L -function, and d^* for the fundamental discriminant of the primitive character inducing χ_d .

Proposition 3.1 (genus factorization of the Epstein L -function). *For the real character χ_2 of $\text{Cl}(\mathcal{O}_n)$, $n \in \{3, 5, 7, 9, 11, 13\}$, and for the character $\chi_{(2,0)}$ at $n = 15$, the representation numbers satisfy, for every $m \geq 1$,*

$$(3.1) \quad R_{\chi_2}(m) = 2 (\text{conv}_{d_1, d_2} * \text{corr})(m), \quad \text{conv}_{d_1, d_2}(m) = \sum_{e|m} \chi_{d_1}(e) \chi_{d_2}(m/e),$$

with the decompositions $d_1 d_2 = -4n^2$ and finite corrections corr (Dirichlet convolution; $\text{corr} = \delta_1$ unless stated) of the table below. Equivalently, as Dirichlet series,

$$L(s, \chi_2) = L(s, \chi_{d_1}) L(s, \chi_{d_2}) C(s), \quad C(s) = \sum_m \text{corr}(m) m^{-s},$$

and consequently

$$(3.2) \quad L'(0, \chi_2) = \frac{2h(d_1^*)}{w(d_1^*)} h(d_2) \log \varepsilon_{d_2} C(0).$$

n	$d_1 \cdot d_2 = -4n^2$	correction $C(s)$	$L'(0, \chi_2)$
3	$(-3) \cdot 12$	—	$\frac{1}{3} \log \varepsilon_{12}$
5	$(-20) \cdot 5$	—	$2 \log \varepsilon_5$
7	$(-7) \cdot 28$	—	$\log \varepsilon_{28}$
9	$(-27) \cdot 12$	$1 + 3^{1-2s}$	$\frac{4}{3} \log \varepsilon_{12}$
11	$(-11) \cdot 44$	—	$\log \varepsilon_{44}$
13	$(-52) \cdot 13$	—	$2 \log \varepsilon_{13}$
15	$(-15) \cdot 60$	—	$4 \log \varepsilon_{60}$

with $\varepsilon_{12} = 2 + \sqrt{3}$, $\varepsilon_5 = \frac{1+\sqrt{5}}{2}$, $\varepsilon_{28} = 8 + 3\sqrt{7}$, $\varepsilon_{44} = 10 + 3\sqrt{11}$, $\varepsilon_{13} = \frac{3+\sqrt{13}}{2}$, $\varepsilon_{60} = 4 + \sqrt{15}$. In every case d_2 is the discriminant of $\mathbb{Q}(\sqrt{n})$: the real quadratic field carrying the genus-character sum at conductor n is $\mathbb{Q}(\sqrt{n})$ itself.

Proof. The identity (3.1) is verified as an identity of integers for all $m \leq 300$ at every level. This is a proof, by Sturm's bound [27]. The left side is the q -expansion of $\frac{1}{2} \sum_{\mathfrak{c}} \chi_2(\mathfrak{c}) \theta_{Q_{\mathfrak{c}}}$, whose constant term vanishes and which, by Hecke and Schoeneberg [22], is a holomorphic modular form of weight 1 on $\Gamma_0(|D|)$, $|D| = 4n^2$, with character χ_D ; the right side is the weight-1 Eisenstein series $E_1(\chi_{d_1}, \chi_{d_2})$ of [2, §4.8], whose constant term vanishes because

both characters are nontrivial, together with its p^a -dilations weighted by the correction coefficients (at $n = 9$: $E + 3E(9\tau)$, E of level 36), a form of the same weight, level and character. A weight-1 form on $\Gamma_0(4n^2)$ vanishing at ∞ to order beyond $\mu(\Gamma_0(4n^2))/12$ — at most 180, attained at $n = 15$, far below the certified 300 — vanishes identically. For the closed form: $L(s, \chi_{d_1})$ differs from the primitive $L(s, \chi_{d_1}^*)$ by Euler factors at primes $p \mid d_1$, $p \nmid d_1^*$, which here occur only at $n = 9$ ($p = 3$, where $\chi_{-3}(3) = 0$ and nothing changes); the positive d_2 is fundamental at every level, $L(0, \chi_{d_2}) = 0$ and $L'(0, \chi_{d_2}) = h(d_2) \log \varepsilon_{d_2}$, while $L(0, \chi_{d_1}^*) = 2h(d_1^*)/w(d_1^*)$ (Dirichlet's class number formulas at $s = 0$). $\square$

At $n = 9$ the factor $C(s) = 1 + 3^{1-2s}$, $C(0) = 4$, together with $d_1^* = -3$ ($w = 6$), gives $\frac{1}{3} \cdot 4 \log \varepsilon_{12}$. This conductor correction at the square conductor — and only there — is thereby proved as an identity; what remains open is its conceptual derivation from the local proper- $\mathcal{O}_9$ -ideal structure at 3, and the general- n law. The composite square-free conductor $n = 15$ needs no correction for its $\mathbb{Q}(\sqrt{15})$ -character, as this reading predicts (verified at 400 digits). All seven closed forms agree with the independent evaluation of Lemma 2.3 to 10^{-249} or better.

Remark 3.2 (the other two real characters at $n = 15$; computational). The characters $\chi_{(0,1)}$ and $\chi_{(2,1)}$ of $\text{Cl}(\mathcal{O}_{15}) \cong \mathbb{Z}/4 \times \mathbb{Z}/2$ admit *no* factorization of the shape (3.1) with a finite correction over the divisor pairs of -900 — an honest negative result of the discovery procedure — but their values are cleanly identified (certified ratio, 398 spare digits at 400 digits):

$$L'(0, \chi_{(0,1)}) = 8 \log \varepsilon_5, \quad L'(0, \chi_{(2,1)}) = \frac{8}{3} \log \varepsilon_{12},$$

the units of the two proper subfield levels 5 and 3. Their factorizations presumably need Euler data at both conductor primes simultaneously; this is an identification, not a proof.

3.2. Euclidean: the j -dressing. For the real character χ_2 split the classes by $\chi_2 = \pm 1$, and let A, B be the products of $j(\mathfrak{c})$ (resp. of $j(\mathfrak{c}) - 1728$) over the two cosets, so that $\Sigma_0(\chi_2) = \log |A/B|$ (resp. $\Sigma_{1728}(\chi_2) = \log |A/B|$). By genus theory A and B lie in the real quadratic subfield $\mathbb{Q}(\sqrt{d_2})$ of the ring class field fixed by $\ker \chi_2$ and the mirror, are algebraic integers, and are conjugate over $\mathbb{Q}$, so A/B has norm 1.

Proposition 3.3 (exact Σ -factorizations; computational). *At every level $n \leq 15$ and for every real character, $A + B$ and AB are certified integers (≥ 174 spare digits at 250 digits, ≥ 315 at 400), $A, B = \frac{s \pm t\sqrt{d_2}}{2}$ in $\mathbb{Q}(\sqrt{n})$, and the norm-one ratio A/B factors exactly, by ideal arithmetic in $\mathbb{Q}(\sqrt{d_2})$ with valuations computed by Hensel lifting and each identity re-multiplied out symbolically,*

$$(3.3) \quad \Sigma_0(\chi_2) = \log \left| \frac{A}{B} \right| = k \log \varepsilon_{d_2} + \sum_p e_p \lambda_p, \quad \lambda_p := \log \left| \frac{\pi_p}{\pi'_p} \right|,$$

over split primes p of $\mathbb{Q}(\sqrt{d_2})$, π_p an explicit generator of $\mathfrak{p}_p^{m_p}$ (m_p the order of $\mathfrak{p}_p$ in the class group, $e_p \in \frac{1}{m_p}\mathbb{Z}$), normalized so that $0 \leq \lambda_p < 2 \log \varepsilon_{d_2}$; and likewise for Σ_{1728} . The split-prime support lies in $\text{GZ}(-4n^2, -3) \cup \{p \mid 2n\}$ for Σ_0 and in $\text{GZ}(-4n^2, -4) \cup \{p \mid 2n\}$ for Σ_{1728} , where $\text{GZ}(D, D')$ is the set of primes dividing some $\frac{|DD'|-x^2}{4}$ [6] — the same Gross–Zagier sets that govern the denominators of the phase in [8][Prop. 6.26]. The certified factorizations, in the canonical normalization:

n	$\Sigma_0(\chi_2)$	$\Sigma_{1728}(\chi_2)$
3	$16 \log \varepsilon_{12} - 3\lambda_{11} - 3\lambda_{23}$	$4 \log \varepsilon_{12} + 2\lambda_{11}$
5	$30 \log \varepsilon_5 - 3\lambda_{11} + 3\lambda_{59} + 3\lambda_{71}$	$28 \log \varepsilon_5 + 2\lambda_{11} + 2\lambda_{19}$
7	$24 \log \varepsilon_{28} - 3\lambda_{47} - 3\lambda_{83} - 3\lambda_{131}$	$18 \log \varepsilon_{28} + 2\lambda_3 - 2\lambda_{19}$
9	$16 \log \varepsilon_{12} + 3\lambda_{11} - 3\lambda_{23} + 3\lambda_{47} + 3\lambda_{179} - 3\lambda_{227} - 3\lambda_{239}$	$24 \log \varepsilon_{12} - 2\lambda_{11} - 4\lambda_{23}$
11	$12 \log \varepsilon_{44} - 3\lambda_{107} + 3\lambda_{167} + 3\lambda_{263} + 3\lambda_{347} - 3\lambda_{359}$	$4 \log \varepsilon_{44} + 6\lambda_{19} + 2\lambda_{43}$
13	$18 \log \varepsilon_{13} - 3\lambda_{23} + 3\lambda_{107} + 3\lambda_{251} - 3\lambda_{311} + 3\lambda_{443} - 3\lambda_{491} + 3\lambda_{503}$	$28 \log \varepsilon_{13} - 2\lambda_3 - 4\lambda_{23} - 2\lambda_{43}$
15 (2, 0)	$42 \log \varepsilon_{60} - 3\lambda_{11} + 3\lambda_{59} + 3\lambda_{71} + 3\lambda_{191} + 3\lambda_{419} + 3\lambda_{479} + 3\lambda_{659}$	$52 \log \varepsilon_{60} + 2\lambda_{11} + 2\lambda_{43} + 2\lambda_{59}$
15 (0, 1)	$132 \log \varepsilon_5 + 3\lambda_{11} + 3\lambda_{59} + 3\lambda_{71} + 3\lambda_{191} - 3\lambda_{419} - 3\lambda_{479} - 3\lambda_{659}$	$124 \log \varepsilon_5 - 2\lambda_{11} - 4\lambda_{19} + 4\lambda_{31} - 2\lambda_{59}$
15 (2, 1)	$62 \log \varepsilon_{12} - 9\lambda_{11} - 6\lambda_{47} - 3\lambda_{59} - 3\lambda_{71} - 3\lambda_{191} + 3\lambda_{419} + 3\lambda_{479} - 3\lambda_{659}$	$16 \log \varepsilon_{12} + 6\lambda_{11} + 4\lambda_{23} + 4\lambda_{47} + 2\lambda_{59}$

(The generators π_p are printed by the script; e.g. at $n = 3$, $\pi_{11} = -4 - \frac{3}{2}\sqrt{12}$, $\pi_{23} = -5 - 2\sqrt{12}$.)

The ± 3 -multiples in Σ_0 and the even coefficients in Σ_{1728} are the $(4, 3)$ -exponent shadow of $u^6 = -\beta^4(\beta - 1728)^3 \cdot (\Delta\text{-quotient})$; note that at $n = 15$ all three real characters carry exact Σ -data although two of them lack the finite genus factorization of their L' . A general law for the exponents k, e_p — a genus-refined Gross–Zagier statement — is open (Section 7).

Theorem 3.4 (closed form for the genus character). *For the real character χ_2 of $\text{Cl}(\mathcal{O}_n)$, $n \leq 13$, and for the $\mathbb{Q}(\sqrt{15})$ -character at $n = 15$,*

$$S(\chi_2) = -2 \frac{2h(d_1^*)}{w(d_1^*)} h(d_2) C(0) \log \varepsilon_{d_2} + \frac{2}{3} \Sigma_0(\chi_2) + \frac{1}{2} \Sigma_{1728}(\chi_2),$$

with the tabulated exact Σ -factorizations: an explicit element of $\mathbb{Q} \log \varepsilon_{\mathbb{Q}(\sqrt{n})} + \sum_p \mathbb{Q} \lambda_p$ over the Gross–Zagier split primes. (The other two real characters at $n = 15$ carry the same Σ -closed forms, with the certified ε -ratio of Remark 3.2 standing in for the L' -factorization.)

Proof. Theorem 2.5 with (3.2) and (3.3). Every ingredient is exact; the assembled identity is additionally checked against the independent evaluation to 10^{-247} . $\square$

So the Euclidean phases are an elliptic-unit system *up to an identified, Gross–Zagier-supported S -integer dressing*: the L' -part is a unit statement, and the j -part is the price of the singular-moduli factors in u^6 .

Remark 3.5 (signs). The factorizations above are of absolute values, and $S(\chi)$ sees only $|u|$. The sign of A/B is also computed exactly by the script. The sign of the phase itself on the ambiguous classes — where it is real — is a separate matter: on the Euclidean side reality is the center criterion of [8][Prop. 7.10], and on the hyperbolic side the sign of u_f on divisor classes is governed by the archimedean law of [8][Thm. 6.28]. Signs play no further role in this paper.

3.3. Hyperbolic genus characters. At $n = 11, 13, 15$ some odd characters of $\text{Cl}(1 - n^2)$ are real. The same machinery evaluates everything.

Proposition 3.6 (hyperbolic genus factorizations). *For the odd real characters of the table, the representation numbers satisfy $R_\chi = 2 \text{conv}_{d_1, d_2}$ exactly (verified for all $m \leq 300$, proved by the Sturm argument of Proposition 3.1, the bounds here being ≤ 32 ; no conductor correction arises for any odd character), and hence*

$$L'(0, \chi) = \frac{2h(d_1^*)}{w(d_1^*)} h(d_2) \log \varepsilon_{d_2} :$$

n	odd χ	$d_1 \cdot d_2 = 1 - n^2$	$L'(0, \chi)$	field
11	$\chi_{(0,1)}$	$(-15) \cdot 8$	$2 \log(1 + \sqrt{2})$	$\mathbb{Q}(\sqrt{2})$
11	$\chi_{(1,0)}$	$(-3) \cdot 40$	$\frac{2}{3} \log(3 + \sqrt{10})$	$\mathbb{Q}(\sqrt{10})$
13	$\chi_{(0,1)}$	$(-7) \cdot 24$	$\log(5 + 2\sqrt{6})$	$\mathbb{Q}(\sqrt{6})$
13	$\chi_{(1,0)}$	$(-8) \cdot 21$	$\log \frac{5 + \sqrt{21}}{2}$	$\mathbb{Q}(\sqrt{21})$
15	$\chi_{(0,1)}$	$(-8) \cdot 28$	$\log(8 + 3\sqrt{7})$	$\mathbb{Q}(\sqrt{7})$
15	$\chi_{(2,1)}$	$(-4) \cdot 56$	$\frac{1}{2} \log(15 + 4\sqrt{14})$	$\mathbb{Q}(\sqrt{14})$

The *even* real characters pair with the complementary fields. At $n = 11$ and $n = 13$ the even characters factor (certified by the same procedure) as

$$L(s, \chi_{-24}) L(s, \chi_5) \quad (n = 11), \quad L(s, \chi_{-3}) L(s, \chi_{56}) \quad (n = 13),$$

so the fields are $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{14})$ — exactly the fields of the pair-sums $u_f + u_f^{-1}$ of [8][Thm. 6.19(3)]. So the parity $\chi(\mathfrak{r}_n) = \pm 1$ splits the genus fields of $1 - n^2$ into the *pair-sum fields* (even; the algebra of $u + 1/u$) and the *log $|u|$ -fields* (odd; the character sums) — complementary halves of one genus theory. At $n = 9$ (disc -80 , conductor 2) the single real character is even, and at $n = 15$ the even character $\chi_{(2,0)}$ appears; their Epstein L -functions factor as

$$L(s, \chi_{-16}) L(s, \chi_5) (1 + 2^{-s} + 2 \cdot 4^{-s}) \quad (n = 9),$$

$$L(s, \chi_{-7}) L(s, \chi_{32}) (1 - 2^{-s} + 2 \cdot 4^{-s}) \quad (n = 15),$$

two more instances of the conductor Euler phenomenon, both certified.

Proposition 3.7 (hyperbolic Σ -factorizations; computational). *For the odd real characters the coset products over $\ker \chi$ are certified conjugate quadratic integers in the matching field, and the ratios factor exactly as in (3.3); where a split prime is non-principal the generator is taken for $\mathfrak{p}^m$ (m the class order) and the exponent is e/m — at $n = 11$ in $\mathbb{Q}(\sqrt{10})$ (class number 2), $\pi_3 = 1 + \frac{1}{2}\sqrt{40}$ generates $\mathfrak{p}_3^2$.*

n	χ	$\Sigma_0(\chi)$	$\Sigma_{1728}(\chi)$
11	$(-15) \cdot 8$	$30 \log \varepsilon_8 + 3\lambda_{41} + 3\lambda_{89}$	$36 \log \varepsilon_8 + 2\lambda_{71}$
11	$(-3) \cdot 40$	$14 \log \varepsilon_{40} + 3\lambda_3 - 3\lambda_{41} - 3\lambda_{89}$	$16 \log \varepsilon_{40} - 2\lambda_{71}$
13	$(-7) \cdot 24$	$18 \log \varepsilon_{24} - 3\lambda_5 + 3\lambda_{101}$	$12 \log \varepsilon_{24} + 4\lambda_{19} + 2\lambda_{47} + 2\lambda_{167}$
13	$(-8) \cdot 21$	$24 \log \varepsilon_{21} - 3\lambda_5 - 3\lambda_{101}$	$24 \log \varepsilon_{21} - 2\lambda_{47} - 2\lambda_{167}$
15	$(-8) \cdot 28$	$30 \log \varepsilon_{28} - 6\lambda_{29} + 3\lambda_{47} - 6\lambda_{53} + 3\lambda_{167}$	$20 \log \varepsilon_{28} + 2\lambda_{31} - 2\lambda_{47} - 2\lambda_{103} - 2\lambda_{199} - 2\lambda_{223}$
15	$(-4) \cdot 56$	$6 \log \varepsilon_{56} - 6\lambda_{11} + 3\lambda_{47} - 3\lambda_{167}$	$2 \log \varepsilon_{56} + 4\lambda_{11} + 2\lambda_{31} + 4\lambda_{43} - 2\lambda_{47} + 2\lambda_{103} - 2\lambda_{199} - 2\lambda_{223}$

The split-prime supports again land in $\text{GZ}(1 - n^2, -3)$ resp. $\text{GZ}(1 - n^2, -4)$ together with $p \mid 2(n^2 - 1)$. Combining with Proposition 3.6 and Theorem 2.9, $S(\chi)$ is in closed form for every odd real character of every computed level, verified to 10^{-247} .

The non-real odd characters. At $n = 9$ ($\text{Cl} \cong \mathbb{Z}/4$) the two odd characters are quartic, and their sums collapse by the twist law to $S(\chi_4) = 2 \log |u_1|$; the master identity is then precisely Lemma 2.8 at the principal class, and $-12L'(0, \chi_4) = \log |R_1|$ is a Stark-type evaluation in the quartic ring class field, certified through the polynomial of Theorem 2.10. The same holds at $n = 15$ for its quartic odd pair.

3.4. Certified non-fits: the Stark regime.

Proposition 3.8 (computational; certified negative results). *For the order-3 and order-6 characters at Euclidean $n = 9, 11, 13$, the order-4 characters at Euclidean $n = 7, 15$, and the odd quartic characters at hyperbolic $n = 9, 15$, an integer-relation search in the information-theoretically safe regime (250 digits, 400 at Euclidean $n = 15$; coefficient height $\leq 10^6$; tolerance 10^{-200} , resp. 10^{-320} ; a relation is accepted only when its information content is at most a quarter of the working precision) finds no rational linear relation among $S(\chi)$, $L'(0, \chi)$, $\{\log p : p \mid 2nH(0)H(1728)\}$ (resp. $p \mid 2(n^2 - 1)H(0)H(1728)$) and $\log \varepsilon_{d_2}$.*

The dressing of a non-real character genuinely lives in the cubic or quartic subfield of the ring class field — its exact description is the certified coset data of Remark 2.7 — and not in the rank-one basis of the naive conjecture. This is the expected Stark-regime behaviour: the master identities are exactly the part that survives, and the recognition of the coset objects as Stark units of the cubic and quartic subfields [20, Ch. 6–7] is the natural continuation (Section 7).

4. THE UNIT THEOREM AND THE PER-CLASS VALUATION LAW

This is the technical core of the paper. Throughout, $n \geq 3$ is odd, $D = 1 - n^2$, $\mathcal{O}$ the order of discriminant D , and the notation of the introduction is in force: $r_0 = \frac{n-1}{2}$, $\mathfrak{r} = [r_0, \omega_0]$, $\mathfrak{b}_1 = [1, m_1]$, $\beta_1 = j(f)$, $\beta_2 = j(\mathfrak{r}f)$, and R_f as in (2.5).

4.1. The lattice form of the twisted Δ -ratio.

Lemma 4.1 (lattice lemma). *For every primitive class f of every odd level $n \geq 3$,*

$$(4.1) \quad R_f = r_0^6 \frac{\Delta(\mathfrak{b}_1)}{\Delta(\mathfrak{r}^{-1}\mathfrak{b}_1)}.$$

In particular the j -dressing of R_f is pure bookkeeping: the twisted ratio is the Siegel-type Δ -quotient along the invertible ambiguous twist, normalized by its norm.

Proof. From the closed form $u_f = -\varepsilon\mu^{-2}h_2(\mathfrak{b}_1)/h_2(\mathfrak{b}_2)$ [8][Prop. 6.10] and the weight algebra $h_2^6 = c j^4(j - 1728)^3\Delta$,

$$R_f = \varepsilon^6 \mu^{-12} \frac{\Delta(\mathfrak{b}_1)}{\Delta(\mathfrak{b}_2)}.$$

Since $[\mathfrak{b}_2] = [\mathfrak{r}^{-1}\mathfrak{b}_1]$ there is $\nu \in K^\times$, unique up to sign, with $\mathfrak{b}_2 = \nu \mathfrak{r}^{-1}\mathfrak{b}_1$ [8][eq. (6.1)]; homogeneity gives

$$\Delta(\mathfrak{b}_2) = \nu^{-12} \Delta(\mathfrak{r}^{-1}\mathfrak{b}_1), \quad \text{so} \quad R_f = \varepsilon^6 \left(\frac{\nu^2}{\mu^2} \right)^6 \frac{\Delta(\mathfrak{b}_1)}{\Delta(\mathfrak{r}^{-1}\mathfrak{b}_1)}.$$

By the theorem $\omega_f = \varepsilon\nu^2/(r_0\mu^2) = 1$ of [8][Thm. 6.18], $\varepsilon\nu^2/\mu^2 = r_0$, whence $\varepsilon^6(\nu^2/\mu^2)^6 = r_0^6$. $\square$

The lemma is verified classwise at every odd $n \leq 21$, with maximal relative deviation $6.5 \cdot 10^{-419}$ at 420 digits, the twist lattice $\mathfrak{r}^{-1}\mathfrak{b}_1$ being computed by exact Hermite-normal-form arithmetic in K . At $n = 3$: $r_0 = 1$, $\mathfrak{r} = \mathcal{O}$, and $R = 1$ exactly — the mandatory anchor.

4.2. The unit theorem.

Theorem 4.2 (unit theorem). *For every odd $n \geq 3$ and every class f of discriminant $1 - n^2$ — primitive or not, with R_f defined for imprimitive classes by the lattice formula (4.1) — one has $v_{\mathfrak{P}}(R_f) = 0$ at every finite place $\mathfrak{P}$ of $\bar{\mathbb{Q}}$: R_f is an algebraic unit. Consequently the level polynomial $\prod_f (x - R_f)$ over the primitive classes is a monic integer polynomial with constant term ± 1 , palindromic by the twist law: the certified observations of Theorem 2.10 are theorems for every n .*

The proof runs through five lemmas. Fix a prime p and a place $\mathfrak{P} \mid p$ of $\bar{\mathbb{Q}}$, with v the valuation normalized by $v(p) = 1$. For a lattice $\mathfrak{a}$ with CM by $\mathcal{O}$ write $E_{\mathfrak{a}}$ for the elliptic curve $\mathbb{C}/\mathfrak{a}$.

Lemma 4.3 (tangent-scalar expression). *Let $\mathfrak{a}$ be a lattice with CM by $\mathcal{O}$, $\mathfrak{c}$ an invertible $\mathcal{O}$ -ideal, and $\varphi : E_{\mathfrak{a}} \rightarrow E_{\mathfrak{c}^{-1}\mathfrak{a}}$ the isogeny that is $z \mapsto z$ analytically. Then*

$$v\left(\frac{\Delta(\mathfrak{a})}{\Delta(\mathfrak{c}^{-1}\mathfrak{a})}\right) = -12v(\delta_{\varphi}),$$

where δ_{φ} is the scalar by which φ pulls back Néron differentials of good-reduction models at $\mathfrak{P}$.

Proof. Choose Weierstrass models (E_i, ω_i) over $\bar{\mathbb{Q}}$ whose period lattices are $\lambda\mathfrak{a}$ and $\mu\mathfrak{c}^{-1}\mathfrak{a}$; then $\Delta(E_1, \omega_1) = \lambda^{-12}\Delta(\mathfrak{a})$, $\Delta(E_2, \omega_2) = \mu^{-12}\Delta(\mathfrak{c}^{-1}\mathfrak{a})$, and $\varphi^*\omega_2 = t\omega_1$ with $t = \mu/\lambda \in \bar{\mathbb{Q}}$ (the isogeny and the differentials are algebraic), so $\Delta(\mathfrak{a})/\Delta(\mathfrak{c}^{-1}\mathfrak{a}) = t^{-12}\Delta(E_1, \omega_1)/\Delta(E_2, \omega_2)$. CM curves have potentially good reduction; over a finite extension $L/\mathbb{Q}_p$ choose minimal models: their discriminants are units, and writing $\omega_i = u_i\omega_{i,\min}$ one gets $v(\Delta(E_i, \omega_i)) = -12v(u_i)$ and $\delta_{\varphi} = t u_1/u_2$; the u_i cancel and the display follows. $\square$

Lemma 4.4 (multiplicativity; prime-to- p triviality). *Tangent scalars multiply along composites, and $v(\delta_{\varphi}) = 0$ whenever $\deg \varphi$ is prime to p .*

Proof. Multiplicativity is the chain rule for pullbacks. Both δ_{φ} and $\delta_{\hat{\varphi}}$ are integral — isogenies extend to Néron models — and $\delta_{\varphi}\delta_{\hat{\varphi}} = \deg \varphi$, a p -adic unit. $\square$

Lemma 4.5 (the p -part of the twist). *$\mathfrak{r} = \mathfrak{b}\mathfrak{c}$ with $\mathfrak{b}, \mathfrak{c}$ invertible $\mathcal{O}$ -ideals, $N\mathfrak{b} = p^k$ where $k = v_p(r_0)$, $N\mathfrak{c}$ prime to p , and $\mathfrak{b}$ inherits both structural properties of $\mathfrak{r}$: $\bar{\mathfrak{b}} = \mathfrak{b}$ and $\mathfrak{b}^2 = (p^k)$.*

Proof. Invertible fractional $\mathcal{O}$ -ideals decompose along primes (localization); let $\mathfrak{b}$ be the invertible ideal with $\mathfrak{b}_q = \mathcal{O}_q$ for $q \neq p$ and $\mathfrak{b}_p = \mathfrak{r}_p$, and $\mathfrak{c} = \mathfrak{b}^{-1}\mathfrak{r}$. Conjugation acts locally, and $\bar{\mathfrak{r}} = \mathfrak{r}$ gives $\bar{\mathfrak{b}} = \mathfrak{b}$. Locally $(\mathfrak{b}^2)_p = (\mathfrak{r}^2)_p = r_0\mathcal{O}_p = p^k\mathcal{O}_p$ and $(\mathfrak{b}^2)_q = \mathcal{O}_q$, which is the local data of the principal ideal (p^k) . $\square$

Lemma 4.6 (rigidity). *The quantity $d_p := v(\delta_{\varphi})$ for the $\mathfrak{b}$ -twist isogeny $\varphi : E_{\mathfrak{a}} \rightarrow E_{\mathfrak{b}^{-1}\mathfrak{a}}$ is the same for every proper $\mathcal{O}$ -ideal $\mathfrak{a}$.*

Proof. $v(\delta_{\varphi})$ is computed on formal groups: it is the tangent valuation of the quotient map $G \rightarrow G/C$ of p -divisible groups over $\mathcal{O}_L$, where $G = E_{\mathfrak{a}}[p^{\infty}]$ (of the good model) and C is the schematic closure of $\ker \varphi = \mathfrak{b}_p^{-1}\mathfrak{a}_p/\mathfrak{a}_p$ — the quotient elliptic curve is the good model of $E_{\mathfrak{b}^{-1}\mathfrak{a}}$ by the Néron property, and Lie of an elliptic curve is Lie of its p -divisible group. So d_p depends only on the isomorphism class of the pair (G, C) . Given two proper ideals, choose an invertible $\mathcal{O}$ -ideal $\mathfrak{q}$ of norm prime to p connecting their classes; the connecting isogeny has degree prime to p , hence induces an isomorphism of p -divisible groups, is $\mathcal{O}_p$ -linear on Tate modules ($T_p E_{\mathfrak{a}} = \mathfrak{a} \otimes \mathbb{Z}_p$, free of rank one over $\mathcal{O}_p$ since proper ideals of quadratic orders are invertible, hence locally principal), and identifies the two kernels $\mathfrak{b}_p^{-1}\mathfrak{a}_p/\mathfrak{a}_p$. Homothety adjusts representatives within a class. $\square$

Proposition 4.7 (balance). $2d_p = k$.

Proof. Compose the $\mathfrak{b}$ -twist at $\mathfrak{a}$ with the $\mathfrak{b}$ -twist at $\mathfrak{b}^{-1}\mathfrak{a}$: since $\mathfrak{b}^2 = (p^k)$, $\mathfrak{b}^{-2}\mathfrak{a} = p^{-k}\mathfrak{a}$, and the composite followed by the isomorphism $z \mapsto p^k z$ is the endomorphism $[p^k]$ of $E_{\mathfrak{a}}$, of tangent scalar p^k . By Lemma 4.4 the tangent valuations add to k (an isomorphism of good models contributes 0); by Lemma 4.6 both steps contribute the same d_p , the twist of a proper ideal being proper. $\square$

Proof of Theorem 4.2. By Lemma 4.1 and Lemma 4.3 applied to the $\mathfrak{r}$ -twist $\varphi : E_{\mathfrak{b}_1} \rightarrow E_{\mathfrak{r}^{-1}\mathfrak{b}_1}$, which by Lemma 4.5 is the composite of the $\mathfrak{b}$ -twist and a twist of degree $N\mathfrak{c}$ prime to p ,

$$v(R_f) = 6v(r_0) - 12v(\delta_\varphi) = 6k - 12d_p = 6k - 6k = 0$$

at every $\mathfrak{P} \mid p$ with $p \mid r_0$ (Lemmas 4.4, 4.6 and Proposition 4.7), and $v(R_f) = 0$ trivially at $p \nmid r_0$, where the whole twist has degree prime to p . For an imprimitive class of content g , $\mathfrak{b}_1$ is proper over the order $\mathcal{O}'$ of discriminant D/g^2 and the twist is by $\mathfrak{r}' := \mathfrak{r}\mathcal{O}'$ — invertible (the extension of an invertible ideal), ambiguous, with $\mathfrak{r}'^2 = r_0\mathcal{O}'$ — so the identical argument runs over $\mathcal{O}'$. Finally the polynomial statement: rationality of $\prod_f (x - R_f)$ is Theorem 2.10; the roots being units, the coefficients are rational algebraic integers, hence integers, with $\prod_f R_f = 1$ for $n \geq 5$ (the twist law pairs f with $\mathfrak{r}f$) and $R = 1$ at $n = 3$; palindromy is the twist law $R_{\mathfrak{r}f} = R_f^{-1}$. $\square$

Remark 4.8 (what the proof did not need). No Kronecker limit formula, no Siegel units, and — despite the Deuring dictionary that motivated the statement — no quasi-canonical-lifting valuation formulas: Gross's theory [5] enters only as the conceptual reading (the twist by an ambiguous ideal of the order is a *horizontal* move, and horizontal moves preserve local Δ -data). The inputs are potentially good reduction, Néron differentials, and the rigidity of p -divisible-group quotients.

4.3. The certified record and the imprimitive strata.

Proposition 4.9 (level polynomials; computational). *For every odd $n \leq 21$ the level polynomial $\prod_{f \text{ prim}} (x - R_f)$ is certified with integer coefficients, palindromic, with constant term 1 (spare digits ≥ 237 at 250 digits for $n \leq 13$, ≥ 275 at 300 digits for $n = 15, 17, 19$, 385 at 420 digits for $n = 21$) — as Theorem 4.2 guarantees — and is moreover irreducible over $\mathbb{Q}$ (exact factorization): the R_f of a level form a single Galois orbit.*

n	h	$\prod_f (x - R_f)$
3	1	$x - 1$
5	2	$x^2 - 34x + 1$
7	2	$x^2 - 2702x + 1$
9	4	$x^4 - 339524x^3 - 95354x^2 - 339524x + 1$
11	4	$x^4 - 56529284x^3 + 1538876166x^2 - 56529284x + 1$
13	4	$x^4 - 11382984004x^3 + 885435408006x^2 - 11382984004x + 1$
15	8	$x^8 - 2628641876392x^7 - 21595933374628x^6 - 1373071731101336x^5 + 9740462908109254x^4 - 1373071731101336x^3 - 21595933374628x^2 - 2628641876392x + 1$
17	4	$x^4 - 673122277718404x^3 + 8553847041196806x^2 - 673122277718404x + 1$
19	8	$x^8 - 186863535844922888x^7 + 44665915402536486036508x^6 + 131633377547326082495944x^5 + 179879784238619113420870x^4 + (\text{sym})$
21	12	$x^{12} - 55348592922774901452x^{11} + 87282798056992201360611266x^{10} + \dots$ (all coefficients printed by the script)

Irreducibility for general n remains open exactly as for the phase polynomials Q_n of [8][Open problem 6.24]: the coincidence set $\{f : R_f = R_1\}$ is a translation-invariant subgroup, and only its triviality is missing.

Proposition 4.10 (imprimitive strata). *Let f be a class of content $g > 1$, $\mathcal{O}'$ the order of discriminant D/g^2 , and $\mathfrak{r}' = \mathfrak{r}\mathcal{O}'$ the induced twist ideal. With R defined by the lattice formula, every stratum polynomial is again an integer palindromic polynomial with constant term ± 1 (Theorem 4.2), and*

$$\mathfrak{r}' \text{ principal in } \mathcal{O}' \implies R \equiv 1 \text{ on the content-}g \text{ stratum.}$$

The converse holds on every computed stratum (computational).

Proof. If $\mathfrak{r}' = (\lambda)$, then $\lambda^2 \mathcal{O}' = \mathfrak{r}'^2 = r_0 \mathcal{O}'$ forces $\lambda^2 = r_0 \zeta$ with $\zeta \in \mathcal{O}'^\times$ a root of unity, so $\Delta(\mathfrak{r}'^{-1} \mathfrak{b}_1) = \lambda^{12} \Delta(\mathfrak{b}_1) = r_0^6 \Delta(\mathfrak{b}_1)$ and $R = 1$. $\square$

n	content	disc	$\mathfrak{r}\mathcal{O}'$	stratum polynomial
7	2	-12	$(\sqrt{-3})$, principal	$x - 1$
7	4	-3	principal	$x - 1$
9	2	-20	(2), principal	$(x - 1)^2$
15	2	-56	$[7, \sqrt{-14}]$, nonprincipal	$x^4 - 1988x^3 - 3194x^2 - 1988x + 1$
17	2	-72	$[4, 6\sqrt{-2}]$, nonprincipal	$x^2 - 9602x + 1$
17	3	-32	$(2\sqrt{-2})$, principal	$(x - 1)^2$
17	6	-8	$(2\sqrt{-2})$, principal	$x - 1$
19	3	-40	(3), principal	$(x - 1)^2$

So nothing breaks on the imprimitive strata: the unit theorem persists; what changes is that the twist can *die* (principal induced ideal), collapsing the stratum to $R \equiv 1$.

4.4. The per-class valuation law. We return to the Euclidean Δ -data $G_c = n^{12} \Delta(\Lambda_c) / \Delta(\mathbb{Z}[i])$ over $\text{Cl}(\mathcal{O}_n)$, with $D_n(x) = \prod_c (x - G_c) \in \mathbb{Z}[x]$ and $|D_n(0)| = |M(n)|$ (Theorem 2.6). The Δ -mass law [8][Thm. 7.18] controls the product of the G_c ; the rigidity of Lemma 4.6 distributes it over the classes.

Theorem 4.11 (per-class valuations). *For every level $n \geq 2$, every prime p with $p^k \parallel n$, and every place $\mathfrak{P} \mid p$ of $\bar{\mathbb{Q}}$ (normalized $v(p) = 1$), the valuation $v_{\mathfrak{P}}(G_{\mathfrak{c}})$ is independent of the class $\mathfrak{c}$, equal to*

$$(4.2) \quad w_p(k) = \begin{cases} 0, & p \text{ split } (p \equiv 1 \pmod{4}), \\ \frac{12(p^k - 1)}{(p-1)p^{k-1}(p+1)}, & p \text{ inert } (p \equiv 3 \pmod{4}), \\ 6(2^k - 1)/2^k, & p = 2; \end{cases}$$

and $v_{\mathfrak{P}}(G_{\mathfrak{c}}) = 0$ at $p \nmid n$.

Proof. Write $\psi_{\mathfrak{c}} : E_{\Lambda_{\mathfrak{c}}} \rightarrow E_0 := E_{\mathbb{Z}[i]}$ for the isogeny $z \mapsto z$ (degree n); by Lemma 4.3, $v(G_{\mathfrak{c}}) = 12k - 12v(\delta)$, where δ is the tangent scalar of the p -primary part of $\psi_{\mathfrak{c}}$ (Lemma 4.4). By duality it suffices to control $\hat{\psi}$ -data on the fixed curve E_0 : the p -primary kernel of the dual is the subgroup $C_{\mathfrak{c}} = p^{-k}\Lambda_{\mathfrak{c},p}/\mathbb{Z}_p[i] \subset E_0[p^{\infty}]$. Local structure: $\Lambda_{\mathfrak{c},p}$ is an invertible $\mathcal{O}_{n,p}$ -module inside $\mathbb{Z}_p[i]$ of index $p^k = [\mathbb{Z}_p[i] : \mathcal{O}_{n,p}]$, hence $\Lambda_{\mathfrak{c},p} = \epsilon \mathcal{O}_{n,p}$ with $\epsilon \in K_p^{\times}$, and the containment and the index force $\epsilon \in \mathbb{Z}_p[i]^{\times}$. Since $\mathbb{Z}_p[i] \subset \text{End}(E_0[p^{\infty}])$ (the closure of $\mathbb{Z}[i]$ in the p -adically complete endomorphism ring), ϵ is an automorphism of the p -divisible group carrying $C_{\mathfrak{c}}$ to $C_{\mathfrak{c}}$: the pairs $(E_0[p^{\infty}], C_{\mathfrak{c}})$ are isomorphic for all classes, so $v(\delta_{\hat{\psi}_{\mathfrak{c}}})$, hence $v(G_{\mathfrak{c}})$, is class-independent. The value is then pinned by the Δ -mass law: $h \cdot w = v_p(\prod_{\mathfrak{c}} G_{\mathfrak{c}}) = v_p(M(n))$, and $v_p(M(n))/h = w_p(k)$ by [8][Thm. 7.18] together with $h = \frac{1}{2}N_e(p^k)N_e(n/p^k)$ [8][Prop. 3.1, Thm. 7.6]. $\square$

Proposition 4.12 (Newton-polygon verification; computational). *The multiset of $\mathfrak{P}$ -adic valuations of the roots of D_n is its Newton polygon at p . For every computed level — $2 \leq n \leq 18, 21, 25, 27, 49$ — and every $p \mid n$, the polygon of the certified D_n (spare digits 140–249; 460 digits at $n = 49$) has a single slope, of the value (4.2), across the full degree h :*

n	polygon slopes	n	polygon slopes
2	2 : 3	12	2 : 9/2; 3 : 3
4	2 : 9/2	14	2 : 3; 7 : 3/2
7	7 : 3/2	16	2 : 45/8
8	2 : 21/4	18	2 : 3; 3 ² : 4
9	3 ² : 4	27	3 ³ : 13/3
11	11 : 1	49	7 ² : 12/7

(split primes: slope 0 throughout, e.g. $n = 5, 13, 25$; at composite n every prime factor is listed).

The denominators of the slopes — $p^{k-1}(p+1)$ at inert p , 2^k at the ramified prime — are exactly the ramification degrees of Gross’s quasi-canonical liftings of level k [5]: the per-class law is the quasi-canonical valuation ladder, obtained here from rigidity plus the elementary mass recursion. We stress that this reading is an interpretation and not an input of the proof.

Corollary 4.13 (the split ladder). *Let $p \equiv 1 \pmod{4}$, $p \nmid 6$, $p^k \parallel n$, $n \neq p^k$. Then for every class, $v_{\mathfrak{p}}(u_c^2) = -4k$.*

Proof. By [8][Lemma 7.12],

$$u^2 = -12\beta(\beta - 1728) \frac{g_2(\Lambda_c)}{g_2(\mathbb{Z}[i])}, \quad \left(\frac{g_2(\Lambda_c)}{g_2(\mathbb{Z}[i])} \right)^3 = \frac{\beta}{1728} \cdot \frac{G_c}{n^{12}},$$

so $v(u^2) = \frac{4}{3}v(\beta) + v(\beta - 1728) + \frac{1}{3}v(G_c) - 4k$. Theorem 4.11 gives $v(G_c) = 0$. At an ordinary place, $\beta \equiv 0$ would force the endomorphism ring of the reduction to contain orders of two distinct imaginary quadratic fields (impossible), and $\beta \equiv 1728$ would force equal prime-to- p conductors, $n/p^k = 1$ (Deuring), which is excluded. $\square$

This proves the exactly arithmetic 5^{4k} -denominator ladder observed at $n = 15$ in [8][§7.4] (with the witness $v_5(H_{-900}(0)) = v_5(H_{-900}(1728)) = 0$ certified). At non-split p the same reduction expresses the fine structure of the normalizers λ_n of [8][§7.4] through $w_p(k)$ plus the β -side collision valuations; the remaining open ingredient is the conductor-degenerate Gross–Zagier data of Section 7, no longer the Δ -part.

Corollary 4.14 (the inert ladders, decomposed; computational). *At $n = 7$ ($p = 7$, $k = 1$) and $n = 9$ ($p = 3$, $k = 2$) every ingredient of $v_{\mathfrak{p}}(u_c^2) = \frac{4}{3}v(\beta) + v(\beta - 1728) + \frac{1}{3}v(G) - 4k$ is certified to be a single Newton-polygon slope — the β -collision valuations are class-independent too — giving the constant per-class values*

$$\begin{aligned} n = 7: \quad v_7(u_c^2) &= \frac{4}{3} \cdot 0 + \frac{1}{4} + \frac{1}{3} \cdot \frac{3}{2} - 4 = -\frac{13}{4}, \\ n = 9: \quad v_3(u_c^2) &= \frac{4}{3} \cdot \frac{1}{2} + \frac{1}{2} + \frac{1}{3} \cdot 4 - 8 = -\frac{11}{2}. \end{aligned}$$

The recorded denominator ladders follow: at $n = 7$ the integrality bound $7^{\lfloor 13k/4 \rfloor} = 7^3, 7^6, 7^9, 7^{13}$ for the elementary symmetric functions e_k of the u_c^2 is met exactly; at $n = 9$ the bound $3^{\lfloor 11k/2 \rfloor}$ is attained at the coset-structured $k = 3, 6$ ($3^{16}, 3^{33}$), with structured cancellation at the other k ($3^2, 3^6, 3^{19}, 3^{22}$).

So the λ_n -question is reduced, at every computed level, to one datum: the per-class β -collision slopes, certified constant here, whose general law is the genus-refined Gross–Zagier problem of Section 7.

5. FIRST-POWER SCHMIDT UNITS

The unit R_f is a sixth-power-level object. A canonical sixth root exists and obeys the laws of the phase at first power. Let $\gamma_2 = E_4/\eta^8$ and $\gamma_3 = E_6/\eta^{12}$ be the canonical holomorphic Weber functions ($\gamma_2^3 = j$, $\gamma_3^2 = j - 1728$), and for a primitive class f let τ_f be the CM point of its reduced form.

Definition 5.1. $w_f := u_f \cdot \frac{(\gamma_2^2 \gamma_3)(\tau_f)}{(\gamma_2^2 \gamma_3)(\tau_f)}$.

Proposition 5.2 (exact first-power laws). *For every odd $n \geq 3$ and every primitive class f :*

- (1) $w_f^6 = R_f$;
- (2) $w_f w_{\tau f} = 1$;
- (3) $\overline{w_f} = w_{f^{-1}}$ whenever neither reduced representative is a boundary form ($b = a$ or $a = c$);
- (4) $\sigma(w_f) = \zeta(\sigma, f) w_{f \circ \sigma}(\sigma)$ with $\zeta(\sigma, f)^6 = 1$, for every $\sigma \in \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$.

Moreover, let τ_0 be the adapted uniformizer of [8]/[Lemma 6.13] for the pair $(\mathbf{b}_1, \mathbf{r}^{-1}\mathbf{b}_1)$; by [8]/[Thm. 6.18],

$$u_f = -\frac{1}{r_0} \frac{j'(\tau_0)}{j'(r_0\tau_0)}.$$

Then

$$(5.1) \quad w_f = -\frac{1}{r_0} \left(\frac{\eta(\tau_0)}{\eta(r_0\tau_0)} \right)^4 \quad \text{up to a sixth root of unity:}$$

a weight-0 η -quotient on $\Gamma_0(r_0)$ evaluated at a Heegner point, well defined up to the μ_6 -valued η^4 -multiplier of the basis choice. The cocycle $\zeta(\sigma, f)$ is this multiplier system.

Proof. (1) $(\gamma_2^2\gamma_3)^6 = j^4(j - 1728)^3$, so $w_f^6 = u_f^6 \beta_2^4(\beta_2 - 1728)^3 / \beta_1^4(\beta_1 - 1728)^3 = R_f$. (2) The twist law $u_f u_{\tau f} = 1$ [8]/[Thm. 6.6] and the telescoping of the γ -ratio along the 2-torsion twist ($\mathbf{r}^2 f = f$, and reduced points are canonical). (3) Off the boundary of the reduced domain $\tau_{f^{-1}} = -\bar{\tau}_f$ exactly, the q -expansions of γ_2, γ_3 are real, so $(\gamma_2^2\gamma_3)(-\bar{\tau}) = \overline{(\gamma_2^2\gamma_3)(\tau)}$, and the mirror law $u_{f^{-1}} = \bar{u}_f$ [8]/[Thm. 6.6] finishes. (4) Sixth powers are strictly equivariant by Theorem 4.2 together with the first-power descent [8]/[Thm. 6.19] (the argument of Theorem 2.10), so the discrepancy at first power is a sixth root of unity. For (5.1): $j' = -2\pi i \gamma_2^2\gamma_3 \eta^4$, and $\tau_f, \tau_{\tau f}$ are $\text{SL}_2(\mathbb{Z})$ -equivalent to $\tau_0, r_0\tau_0$, under which $\gamma_2^2\gamma_3$ picks up sixth roots of unity. $\square$

Proposition 5.3 (the coherence table; computational). *Let $m(n) \mid 6$ be minimal with $\{w_f^m\}$ Galois-stable (equivalently, $\prod_f (x - w_f^m)$ a certified integer polynomial). For all odd $3 \leq n \leq 35$:*

n	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31	33	35
$m(n)$	1	3	1	3	6	1	1	3	2	3	6	1	6	6	6	6	6

At the fully coherent levels the first-power Schmidt units exist unconditionally, with certified irreducible minimal polynomials

$$\begin{aligned}
n = 3: & \quad x + 1 \quad (w = -1), \\
n = 7: & \quad x^2 + 4x + 1 \quad (w = -\varepsilon_{12}^{\mp 1} = -2 \pm \sqrt{3}), \\
n = 13: & \quad x^4 + 50x^3 + 123x^2 + 50x + 1, \\
n = 15: & \quad x^8 + 112x^7 - 630x^6 + 1568x^5 - 2109x^4 \\
& \quad + 1568x^3 - 630x^2 + 112x + 1,
\end{aligned}$$

$$n = 25 : \quad x^8 + 13346x^7 + 24067x^6 - 4876x^5 + 11653x^4 \\ - 4876x^3 + 24067x^2 + 13346x + 1.$$

The case $n = 7$ is emblematic: the phase of a level-7 Schmidt circle, γ -normalized, *is* the fundamental unit of $\mathbb{Q}(\sqrt{3})$ up to sign. At the partially coherent levels the intermediate polynomials are equally clean: for w^3 , $x^2 + 6x + 1$ at $n = 5$ ($w^3 = -\varepsilon_8^{\pm 2}$), $x^4 + 584x^3 + 766x^2 + 584x + 1$ at $n = 9$, $x^4 + 25944604x^3 + 99499206x^2 + 25944604x + 1$ at $n = 17$; for w^2 at $n = 19$, $x^8 - 571772x^7 + 36149386x^6 - 42809072x^5 + 82773139x^4 - (\text{sym})$.

The pattern of the table resists a congruence law: the sign part m_2 and the cube part m_3 of $m(n)$ vary independently: in the computed range $m_2 = 2$ exactly at $n = 11, 19, 23, 27, 29, 31, 33, 35$, and $m_3 = 3$ exactly at $n = 5, 9, 11, 17, 21, 23, 27, 29, 31, 33, 35$; and every candidate congruence law formulated on the data through $n = 27$ was falsified at $29 \leq n \leq 35$. The mechanism is visible at the boundary forms, where the mirror law genuinely fails by the explicit factors

$$(\gamma_2^2 \gamma_3)(\tau + 1) = -\zeta_3 (\gamma_2^2 \gamma_3)(\tau), \quad (\gamma_2^2 \gamma_3)(-1/\tau) = -(\gamma_2^2 \gamma_3)(\tau)$$

(non-real w on ambiguous boundary classes at $n = 31$); re-evaluating the γ 's on a Schertz 6-system, or using the η -quotient form with a Hermite-adapted τ_0 , moves the failure elsewhere without removing it.

Open problem 5.4 (the cocycle). Determine $\zeta(\sigma, f)$ — the multiplier book-keeping of Kubert–Lang and Schertz [10, Ch. 11], [20] for the hybrid object w_f , a class function u_f times a ratio of level-6 class invariants — and derive $m(n)$. The certified table of Proposition 5.3 is the constraint any such computation must reproduce; the η^4 -form (5.1) suggests a Dedekind-sum formula in (r_0, s_0) .

6. THE ROBERT INDEX

6.1. The quadratic layer.

Proposition 6.1 (eigenprojections against fundamental units). *Let χ be an odd real character of $\text{Cl}(1 - n^2)$ with proved factorization $L(s, \chi) = L(s, \chi_{d_1})L(s, \chi_{d_2})C(s)$ (Proposition 3.6), or a real character of $\text{Cl}(\mathcal{O}_n)$ with the factorization of Proposition 3.1, and set*

$$m_\chi := \frac{2h(d_1^*)}{w(d_1^*)} h(d_2) C(0) \in \mathbb{Q}.$$

Then

$$\sum_f \chi(f) \log |R_f| = -24 m_\chi \log \varepsilon_{d_2} \quad (\text{hyperbolic}), \\ \sum_{\mathfrak{c}} \chi(\mathfrak{c}) \log |G_{\mathfrak{c}}| = -12 m_\chi \log \varepsilon_{d_2} \quad (\text{Euclidean}):$$

the χ -eigenprojection of the Schmidt-unit system against the fundamental unit of its real quadratic field is an explicit integer multiple of $\log \varepsilon_{d_2}$ built from the class numbers $h(d_1^*)$, $h(d_2)$ and the conductor multiplier $C(0)$.

Proof. Theorem 2.10, resp. Theorem 2.6(2), with (3.2). $\square$

The certified values (residuals $\leq 10^{-248}$ at 250 digits):

aspect	n	χ	projection
hyperbolic	11	(0, 1)	$-48 \log \varepsilon_8$
hyperbolic	11	(1, 0)	$-16 \log \varepsilon_{40}$
hyperbolic	13	(0, 1), (1, 0)	$-24 \log \varepsilon_{24}, -24 \log \varepsilon_{21}$
hyperbolic	15	(0, 1), (2, 1)	$-24 \log \varepsilon_{28}, -12 \log \varepsilon_{56}$
Euclidean	3, 5, 7	χ_2	$-4 \log \varepsilon_{12}, -24 \log \varepsilon_5, -12 \log \varepsilon_{28}$
Euclidean	9, 11, 13	χ_2	$-16 \log \varepsilon_{12}, -12 \log \varepsilon_{44}, -24 \log \varepsilon_{13}$

6.2. The cubic layer: the index is $8h_{L_3}$. Among the Euclidean levels $n \leq 31$, the ring class group $\text{Cl}(\mathcal{O}_n)$ has 3-torsion exactly at $n = 9, 11, 13, 18, 22, 23, 26, 27$. At such a level let χ_3 be a cubic character, F the cyclic cubic extension of K inside H_n cut out by $\ker \chi_3$ — a dihedral sextic over $\mathbb{Q}$ — and $L_3 \subset F$ its real cubic subfield, the fixed field of $\ker \chi_3$ and the mirror. Let

$$\theta := \prod_{\mathfrak{c} \in \ker \chi_3} G_{\mathfrak{c}}$$

be the principal Δ -mass coset product — at $n = 9, 11, 13$ the real root of the coset cubics of Remark 2.7. Since $v_p(M(n))$ is a multiple of 3 at every p (the exponents of [8][Thm. 7.18] carry the factor $6/e_p \in \{3, 6\}$), $|M(n)|^{1/3}$ is an integer: $3^8, 11^2, 1, 2^{12}3^{16}, 2^{12}11^4, 23^2, 2^{12}, 3^{26}$ at the eight levels.

Proposition 6.2. *The number $\theta_u := \theta/|M(n)|^{1/3}$ is an algebraic unit of L_3 , and $\log |\theta_u| = -8L'(0, \chi_3)$ at the real embedding.*

Proof. θ is fixed by the translations in $\ker \chi_3$ (Theorem 2.6(1)) and by complex conjugation (which permutes the coset $\ker \chi_3$), so $\theta \in L_3$, real. Integrality both ways: by Theorem 4.11 the valuation of every $G_{\mathfrak{c}}$ at every place over $p \mid n$ is the constant $w_p(k)$ and 0 elsewhere, so $v(\theta) = \frac{h}{3}w_p(k) = \frac{1}{3}v(M(n))$ over $p \mid n$ and 0 elsewhere — exactly the valuations of $|M(n)|^{1/3}$. For the logarithm write $a = \log |\theta|$ and b for the common log-modulus of the two conjugate coset products (they are complex conjugates of each other, the mirror exchanging the two nontrivial cosets). The uniform Stark law $-12L'(0, \chi_3) = \sum \chi_3 \log |G|$ of Theorem 2.6(2) reads $a - b = -12L'$ (as $\zeta_3 + \zeta_3^2 = -1$), while $a + 2b = \log |M(n)|$; solving, $\log |\theta_u| = a - \frac{1}{3} \log |M(n)| = \frac{2}{3}(a - b) = -8L'(0, \chi_3)$. $\square$

Lemma 6.3 (conductor Euler factors). *Let χ be a character of $\text{Cl}(\mathcal{O}_n)$. The Epstein L -function is the sum $L(s, \chi) = \sum_{\mathfrak{b}} \bar{\chi}(\mathfrak{b}) N(\mathfrak{b})^{-s}$ over all invertible integral $\mathcal{O}_n$ -ideals, and factors as $L(s, \chi) = L_{\text{prim}}(s, \chi) C_n(s)$, where $L_{\text{prim}}(s, \chi)$ is the Hecke L -function of χ as a primitive ring class character*

of K and $C_n(s) = \prod_{p|n} C_{n,p}(s)$ collects the Euler factors of the conductor-supported invertible ideals. If χ is nontrivial on $\ker(\mathrm{Cl}(\mathcal{O}_n) \rightarrow \mathrm{Cl}(\mathcal{O}_{n/p}))$ then $C_{n,p} \equiv 1$; in particular $C_n \equiv 1$ when χ factors through no $\mathrm{Cl}(\mathcal{O}_m)$, $m \mid n$, $m < n$.

Proof. For an invertible $\mathfrak{a}$, $\alpha \mapsto (\alpha)\mathfrak{a}^{-1}$ bijects the nonzero elements of $\mathfrak{a}$ modulo ± 1 with the invertible integral ideals in the class of $\mathfrak{a}^{-1}$, and $N(\alpha)/N\mathfrak{a} = N((\alpha)\mathfrak{a}^{-1})$; this gives the ideal sum, and the Euler product over $p \nmid n$ is the usual one. At $p \mid n$ the local order $\mathcal{O}_{n,p} = \mathbb{Z}_p + p^k \mathcal{O}_{K,p}$ is local with maximal ideal $p\mathbb{Z}_p + p^k \mathcal{O}_{K,p}$, and its integral invertible ideals are the $x\mathcal{O}_{n,p}$, $x \in \mathcal{O}_{n,p} \setminus \{0\}$ modulo $\mathcal{O}_{n,p}^\times$. For a non-unit x one checks $x\mathcal{O}_{n/p,p}^\times \subseteq \mathcal{O}_{n,p}$, so the group $U' = \mathcal{O}_{n/p,p}^\times / \mathcal{O}_{n,p}^\times$ acts freely on the non-unit ideals, preserving norms; the class of $u\mathcal{O}_{n,p}$ ($u \in U'$, extended by $\mathcal{O}_{n,q}$ at $q \neq p$) runs through $\ker(\mathrm{Cl}(\mathcal{O}_n) \rightarrow \mathrm{Cl}(\mathcal{O}_{n/p}))$, and χ is multiplicative along the orbit, so every orbit sum carries the factor $\sum_{u \in U'} \bar{\chi}(u\mathcal{O}_n) = 0$. Only the unit ideal survives: $C_{n,p} \equiv 1$. $\square$

Proposition 6.4. L_3 has one real and one complex place (unit rank 1), and

$$h_{L_3} R_{L_3} C_n(0) = L'(0, \chi_3),$$

where $L(s, \chi_3)$ is the Epstein L -function of level n and $C_n(s)$ the finite Euler product of Lemma 6.3, with $C_n \equiv 1$ at $n = 9, 11, 13, 23$, where χ_3 is primitive (the class groups of the proper divisor levels have no 3-torsion).

Proof. $\zeta_{L_3}(s) = \zeta(s)L_{\mathrm{prim}}(s, \chi_3)$ by the Artin formalism for the dihedral extension $F/\mathbb{Q}$ ($\zeta_{L_3} = \zeta \cdot L(s, \rho)$ with $\rho = \mathrm{Ind}_K^{\mathbb{Q}} \chi_3$ the two-dimensional representation, and $L(s, \rho) = L_{\mathrm{prim}}(s, \chi_3)$), and $L(s, \chi_3) = L_{\mathrm{prim}}(s, \chi_3)C_n(s)$ by Lemma 6.3. Since $L(0, \chi_3) = 0$, $L'(0, \chi_3) \neq 0$ (Proposition 2.2 and the nonvanishing of $L(1, \chi_3)$) and $\zeta(0) = -\frac{1}{2}$, the function ζ_{L_3} vanishes to first order at $s = 0$: $r_1 + r_2 - 1 = 1$ with a real embedding present, forcing $(r_1, r_2) = (1, 1)$. The class number formula at $s = 0$ [16, Cor. VII.5.11], $\zeta'_{L_3}(0) = -h_L R_L / w_L$ with $w_L = 2$, gives $h_L R_L = -2\zeta'_{L_3}(0) = L'_{\mathrm{prim}}(0, \chi_3)$, and $L'(0, \chi_3) = C_n(0)L'_{\mathrm{prim}}(0, \chi_3)$ since $L_{\mathrm{prim}}(0, \chi_3) = 0$. $\square$

Lemma 6.5 (fundamental units; computational, rigorous). *For each of the eight levels, the real root η_L of the polynomial listed in Theorem 6.6 is a fundamental unit of L_3 , $R_{L_3} = \log |\eta_L|$. Method: starting from θ_u , every k -th root, $2 \leq k \leq 40$, is tested for certified integrality of its minimal polynomial (absolute-error criterion, ≥ 249 spare digits at 250), descending whenever a root lies in L_3 ; the descent terminates at η_L . If η_L were a proper power η_{true}^J , $J > 1$, then some prime $k \mid J$ would satisfy $k \leq J = R_{L_3}/R_{\mathrm{true}} < R_{L_3}/0.2052 < 40$ by Friedman's unconditional bound $R_K > 0.2052$ for every number field $K \neq \mathbb{Q}$ [3], and $\eta_L^{1/k} \in L_3$ would have been detected. The largest ratio is $R_{L_3}/0.2052 < 31.2$ (at $n = 23$).*

Theorem 6.6 (Robert index, cubic layer). *At every Euclidean level with a cubic character, $n \in \{9, 11, 13, 18, 22, 23, 26, 27\}$,*

$$[\mathcal{O}_{L_3}^\times : \langle -1, \theta_u \rangle] = \frac{\log |\theta_u|}{R_{L_3}} = 8 \frac{L'(0, \chi_3)}{R_{L_3}} = 8 h_{L_3} C_n(0),$$

where $C_n(0) = 1$ when χ_3 is primitive and $C_n(0)$ is the integer imprimitivity multiplier of Lemma 6.3 otherwise. The identity is Propositions 6.2–6.4; the content of the theorem is the certified computation of the fundamental units (Lemma 6.5), hence of R_{L_3} and of the integer $L'(0, \chi_3)/R_{L_3} = h_{L_3} C_n(0)$ (≥ 249 spare digits, L' by the independent evaluation of Lemma 2.3):

n	χ_3	fundamental unit of L_3	R_{L_3}	h_{L_3}	$C_n(0)$	index
9	primitive	$x^3 + 15x^2 + 57x - 1$	4.0476417264	1	1	8
11	primitive	$x^3 - 25x^2 + 201x - 1$	5.3026856598	1	1	8
13	primitive	$x^3 - x^2 + 9x - 1$	2.1860813593	3	1	24
23	primitive	$x^3 - 49x^2 + 601x - 1$	6.3984592530	2	1	16
18	from $n = 9$	$x^3 + 15x^2 + 57x - 1$	4.0476417264	1	2	16
22	from $n = 11$	$x^3 - 25x^2 + 201x - 1$	5.3026856598	1	2	16
26	from $n = 13$	$x^3 - x^2 + 9x - 1$	2.1860813593	3	2	48
27	from $n = 9$	$x^3 + 15x^2 + 57x - 1$	4.0476417264	1	4	32

The minimal polynomials of θ_u (constant term -1 ; the regulators are shown to 10 decimals, and are certified to 250):

$$\begin{aligned}
n = 9: & \quad x^3 - 11708931x^2 + 115597311109635x - 1, \\
n = 11: & \quad x^3 + 2297078781x^2 + 2651044211389651971x - 1, \\
n = 13: & \quad x^3 + 446643445245x^2 + 61048319249786206560771x - 1, \\
n = 23: & \quad x^3 - 28994720086003708422147x^2 \\
& \quad + 289100431655153833437856247634098764691761155x - 1, \\
n = 18: & \quad x^3 + 94095557056509x^2 + 13362738335777743394966415363x - 1, \\
n = 22: & \quad x^3 + 25517496658857981x^2 + 7028035410742581725200408425098342403x - 1, \\
n = 26: & \quad x^3 - 77393728680750899988483x^2 \\
& \quad + 3726897283223817102236830775918158257403004931x - 1, \\
n = 27: & \quad x^3 + 17871502813780746125713563645x^2 \\
& \quad + 178562775830464135183026414005799966775800870888956534787x - 1.
\end{aligned}$$

Proof. $\mathcal{O}_{L_3}^\times = \{\pm 1\} \times \langle \eta_L \rangle$ by Proposition 6.4 and Lemma 6.5, so the index of $\langle -1, \theta_u \rangle$ is $\log |\theta_u| / \log |\eta_L|$, which is $8L'(0, \chi_3)/R_{L_3}$ by Proposition 6.2 and $8h_{L_3}C_n(0)$ by Proposition 6.4. At the primitive levels h_{L_3} is the certified integer $L'(0, \chi_3)/R_{L_3}$; at the pullback levels the cubic field, hence h_{L_3} and R_{L_3} , is that of the primitive level, and the certified integer $L'(0, \chi_3)/R_{L_3}$ is $h_{L_3}C_n(0)$, from which $C_n(0)$ is read off. $\square$

Three readings. (i) At $n = 13$ and $n = 23$ the class numbers $h_{L_3} = 3, 2$ of the cubic fields are forced into the index — precisely the Robert–Kubert–Lang phenomenon, “cyclotomic units have index h^+ ”, realized by

the Schmidt disks of curvature $2n$. The row $n = 23$ is a genuine out-of-sample confirmation: a new cubic field, computed after the law $8h_{L_3}$ had been formulated on 9, 11, 13. (ii) The pullback levels are a strong internal consistency check: the *identical* fundamental units and regulators recur at $n = 9, 18, 27$, at $n = 11, 22$ and at $n = 13, 26$, while the coset units θ_u differ, and the index grows by exactly the local Euler value $C_n(0) - 2$ per added prime, 4 at the square-conductor step $9 \rightarrow 27$ — the same local Euler phenomenon as the conductor corrections of Section 3. (iii) The universal factor $8 = 12/(3/2)$ is the weight bookkeeping of the Δ -normalization: the coset object is a 12th-power-level datum, projected with the $\frac{3}{2}$ -coefficient of the Stark relation of Remark 2.7.

6.3. The full index. The cubic layer is one eigen-piece of a single object. Let

$$\mathcal{V}_n := \langle G_c/G_{c'} : c, c' \in \text{Cl}(\mathcal{O}_n) \rangle = \langle G_c/G_1 : c \neq 1 \rangle \subset \mathcal{O}_{H_n}^\times$$

be the group of ratios of the Δ -data: by Theorem 4.11 every ratio is a unit, and by Theorem 2.6(1) the group is stable under $\text{Gal}(H_n/\mathbb{Q})$. $H_n \supset K$ is totally complex of degree $2h$, generalized dihedral over $\mathbb{Q}$, of unit rank $h - 1$. For a nontrivial character χ of $\text{Cl}(\mathcal{O}_n)$ write $L(s, \chi) = C_\chi(s) L_{\text{prim}}(s, \chi)$ as in Lemma 6.3, so that $C_\chi(0) = 1$ when χ factors through no $\text{Cl}(\mathcal{O}_m)$, $m \mid n$, $m < n$.

Theorem 6.7 (the full Robert index). *For every level $n \geq 2$ the group $\mathcal{V}_n$ is free abelian of rank $h - 1$, meets $\mu(H_n)$ trivially, and*

$$[\mathcal{O}_{H_n}^\times : \mu(H_n) \mathcal{V}_n] = \frac{24^{h-1} \prod_{\chi \neq 1} |L'(0, \chi)|}{R_{H_n}} = \frac{4 \cdot 24^{h-1}}{w_{H_n}} h_{H_n} \prod_{\chi \neq 1} C_\chi(0),$$

where $\prod_{\chi \neq 1} C_\chi(0)$ is a positive integer and $w_{H_n} = |\mu(H_n)| = 4 \cdot 2^{[4|n]} \cdot 3^{[3|n]}$.

The proof occupies the rest of this subsection; it has four ingredients.

Lemma 6.8 (Dedekind's group determinant; [29, Lemma 5.26]). *For a finite abelian group G and any $g : G \rightarrow \mathbb{C}$, $\det(g(\sigma\tau^{-1}) - g(\sigma))_{\sigma, \tau \neq 1} = \prod_{\chi \neq 1} \sum_{\sigma \in G} \chi(\sigma) g(\sigma)$.*

Proof. The full matrix $A = (g(\sigma\tau^{-1}))_{\sigma, \tau}$ has the characters as eigenvectors with eigenvalues $S_\chi(g) = \sum_{\sigma} \chi(\sigma) g(\sigma)$, so $\det A = \prod_{\chi} S_\chi(g)$. Subtracting the column $\tau = 1$ from the others and then replacing the row $\sigma = 1$ by the sum of all rows turns A into a matrix whose first row is $(S_1(g), 0, \dots, 0)$ and whose complementary minor is the matrix of the lemma; hence $\det A = S_1(g) \cdot \det(\dots)$, and the identity follows (as a polynomial identity in the values of g). $\square$

Lemma 6.9 (primitive level = conductor). *Let $p \mid n$ and let χ be a character of $\text{Cl}(\mathcal{O}_n)$ whose conductor, as a Hecke character on $I_K(n)$, divides $\frac{n}{p} \mathcal{O}_K$. Then χ factors through $\text{Cl}(\mathcal{O}_{n/p})$. Consequently the levels $m \mid n$ through*

which χ factors are closed under gcd, χ has a smallest such level m_χ , and $L(s, \chi_{m_\chi}) = L_{\text{prim}}(s, \chi)$.

Proof. In the description $\text{Cl}(\mathcal{O}_m) = I_K(m)/P_{K,\mathbb{Z}}(m)$ [1, Prop. 7.22], with $P_{K,\mathbb{Z}}(m)$ the principal ideals (α) , $\alpha \equiv a \pmod{m\mathcal{O}_K}$, $a \in \mathbb{Z}$ prime to m , the projection $\text{Cl}(\mathcal{O}_n) \rightarrow \text{Cl}(\mathcal{O}_{n/p})$ is induced by $I_K(n) \subset I_K(n/p)$. Let $(\gamma) \in P_{K,\mathbb{Z}}(n/p) \cap I_K(n)$, $\gamma \equiv c \pmod{\frac{n}{p}\mathcal{O}_K}$; changing c by a multiple of n/p we may assume $\gcd(c, n) = 1$. Then $(\gamma) = (\gamma/c)(c)$ with $(c) \in P_{K,\mathbb{Z}}(n)$, on which χ is trivial, and $\gamma/c \equiv 1 \pmod{\times \frac{n}{p}\mathcal{O}_K}$, so (γ/c) lies in the ray group of the conductor, on which χ is trivial. Hence χ is trivial on $P_{K,\mathbb{Z}}(n/p) \cap I_K(n)$, the kernel of the projection. The gcd-closure is the same argument with $P_{K,1}(\gcd(m_1, m_2)) = P_{K,1}(m_1)P_{K,1}(m_2)$. At the level m_χ every $p \mid m_\chi$ has a prime above it dividing the conductor, hence (conjugation-symmetry of the conductor of a ring class character) all primes above p ; so the ideals prime to the conductor are the ideals prime to m_χ , and Lemma 6.3 gives $L(s, \chi_{m_\chi}) = L_{\text{prim}}(s, \chi)$. $\square$

Lemma 6.10 (the Hecke recursion as an identity of Dirichlet series). *Let $\chi \neq 1$ be a character of $\text{Cl}(\mathcal{O}_n)$, ℓ a prime, and $\chi^{(n\ell)}$ its pullback. Then*

$$L(s, \chi^{(n\ell)}) = \begin{cases} (1 + \ell^{1-2s} - \ell^{-s}(\chi(\mathfrak{l}) + \bar{\chi}(\mathfrak{l})))L(s, \chi) & \ell = \bar{\ell}, \ell \nmid n, \\ (1 + \ell^{1-2s})L(s, \chi) & \ell \text{ inert}, \ell \nmid n, \\ (1 + 2^{1-2s} - 2^{-s}\chi(\mathfrak{p}))L(s, \chi) & \ell = 2 \nmid n, \\ (1 + \ell^{1-2s})L(s, \chi) - \ell^{-2s}m_\pi L(s, \chi^{(n/\ell)}) & \ell \mid n, \end{cases}$$

the last term present only if χ factors through $\text{Cl}(\mathcal{O}_{n/\ell})$, $m_\pi = |\ker(\text{Cl}(\mathcal{O}_n) \rightarrow \text{Cl}(\mathcal{O}_{n/\ell}))|$. Consequently $C_\chi(s)$ is a product of polynomials in ℓ^{-s} , and $C_\chi(0)$ is a product over $\ell \mid n/m_\chi$ of terms $a_{\ell,k}$ defined by $a_{\ell,0} = 1$, $a_{\ell,1} = \ell + 1 - \chi(\mathfrak{l}) - \bar{\chi}(\mathfrak{l})$, $\ell + 1$, $3 - \chi(\mathfrak{p})$ or $\ell + 1$ in the four cases, $a_{\ell,k+1} = (\ell + 1)a_{\ell,k} - m_{\ell,k}a_{\ell,k-1}$ with $m_{\ell,1} = \ell - \chi_{-4}(\ell)$ ($\ell \nmid m_\chi$), ℓ ($\ell \mid m_\chi$), and $m_{\ell,k} = \ell$ for $k \geq 2$. Every $a_{\ell,k}$ ($k \geq 1$) is a real algebraic integer ≥ 2 ; hence $C_\chi(0) > 0$, $L'(0, \chi) = C_\chi(0)L'_{\text{prim}}(0, \chi) \neq 0$, and $\prod_{\chi \neq 1} C_\chi(0)$ is a positive rational integer.

Proof. The form of a class is $x \mapsto N(x)$ on its lattice, so $\zeta_{\mathfrak{c}}(s) = \sum'_{x \in \Lambda_{\mathfrak{c}}} N(x)^{-s}$. Sum the theta series over the $\ell + 1$ sublattices of index ℓ of $\Lambda_{\mathfrak{c}}$: a point lies in $\ell + 1$ of them if it is in $\ell\Lambda_{\mathfrak{c}}$ and in one otherwise, giving $(1 + \ell^{1-2s})\zeta_{\mathfrak{c}}(s)$; the $\mathcal{O}_n$ -stable sublattices contribute $\ell^{-s}\zeta_{[\mathfrak{l}]\mathfrak{c}}$ (resp. $\ell^{-2s}\zeta_{\pi(\mathfrak{c})}^{(n/\ell)}$ for $\ell \mid n$) and the others are the primitive representatives of the fiber over $\mathfrak{c}$ (the fiber lemma of the Euler-system document accompanying this paper); sum against $\frac{1}{2}\chi(\mathfrak{c})$. Differentiating at $s = 0$, where every L vanishes, gives the recursion for the $L'(0, \cdot)$ along a chain of pullbacks from the primitive level, where $L = L_{\text{prim}}$ (Lemma 6.9). Positivity: $a_{\ell,1} \geq \ell - 1 \geq 2$ in every case ($|\chi(\mathfrak{l})| = 1$; $[\mathfrak{p}_n]^2 = 1$ so $\chi(\mathfrak{p}) = \pm 1$); $a_{\ell,2} - a_{\ell,1} = \ell a_{\ell,1} - m_{\ell,1} > 0$; and $a_{\ell,k+1} - a_{\ell,k} = \ell(a_{\ell,k} - a_{\ell,k-1})$ for $k \geq 2$. Nonvanishing of $L'_{\text{prim}}(0, \chi)$ is Hecke's $L_{\text{prim}}(1, \chi) \neq 0$ through the functional equation; Galois invariance of the product gives integrality. $\square$

Lemma 6.11 (roots of unity). $\mu(H_n) = \mu_w$ with $w = 4 \cdot 2^{[4|n]} \cdot 3^{[3|n]}$.

Proof. The maximal subfield of H_n abelian over $\mathbb{Q}$ has exponent-2 Galois group, so $\zeta_m \in H_n$ forces $m \mid 24$; $i \in H_n$; and a biquadratic $K(\sqrt{d})$ lies in the ring class field H_n iff its conductor over K divides $n\mathcal{O}_K$ [1, Thm. 9.18]: the conductor of $K(\sqrt{-3})/K$ is (3), that of $K(\sqrt{2}) = \mathbb{Q}(\zeta_8)$ is $(1+i)^4 = (4)$. $\square$

Proof of Theorem 6.7. The archimedean places of H_n are the conjugate pairs $\{\sigma_{\mathfrak{a}}, \iota\sigma_{\mathfrak{a}}\}$, $\mathfrak{a} \in \text{Cl}(\mathcal{O}_n)$, where $\sigma_{\mathfrak{a}}$ is the Artin automorphism, extended to the fixed complex embedding in which the $G_{\mathfrak{c}}$ are computed; in the normalization $|x|_v = |\tau x|^2$ of a complex place, Theorem 2.6(1) gives $\log |G_{\mathfrak{c}}/G_1|_{v_{\mathfrak{a}}} = 2(f(\mathfrak{a}^{-1}\mathfrak{c}) - f(\mathfrak{a}^{-1}))$ with $f(\mathfrak{c}) := \log |G_{\mathfrak{c}}|$. For a subgroup $U \subset \mathcal{O}_F^{\times}$ generated by $r = r_1 + r_2 - 1$ units, the determinant $R(U)$ of the $r \times r$ matrix of their logarithms at any r of the places is the covolume of the image of U in the unit lattice in the normalization of the regulator, so if $R(U) \neq 0$ then U is free of rank r , $U \cap \mu(F) = 1$ and $[\mathcal{O}_F^{\times} : \mu(F)U] = R(U)/R_F$ [29, Lemma 4.15]. Omitting the place of the identity, the matrix for $U = \mathcal{V}_n$ is $M = (2(g(\mathfrak{a}\mathfrak{c}^{-1}) - g(\mathfrak{a})))_{\mathfrak{a}, \mathfrak{c} \neq 1}$ with $g(x) = f(x^{-1})$ (the group is abelian), which is the matrix of Lemma 6.8: $\det M = 2^{h-1} \prod_{\chi \neq 1} S_{\bar{\chi}}(f) = 2^{h-1} \prod_{\chi \neq 1} S_{\chi}(f)$, and $S_{\chi}(f) = -12L'(0, \chi)$ by Theorem 2.6(2), real since $f(\mathfrak{c}^{-1}) = f(\mathfrak{c})$. So $R(\mathcal{V}_n) = 24^{h-1} \prod_{\chi \neq 1} |L'(0, \chi)| \neq 0$ by Lemma 6.10, which gives the rank, the torsion-freeness and the first formula. For the second, $\zeta_{H_n}(s) = \zeta_K(s) \prod_{\chi \neq 1} L_{\text{prim}}(s, \chi)$ (Artin formalism), and comparing the coefficients of s^{h-1} in Dedekind's $\zeta_F(s) \sim -\frac{h_F R_F}{w_F} s^{r_F}$ [16, Cor. VII.5.11] with $\zeta_K(0) = -\frac{1}{4}$ and the simple zeros of the $L_{\text{prim}}(s, \chi)$ gives $\prod_{\chi \neq 1} L'_{\text{prim}}(0, \chi) = 4h_{H_n} R_{H_n}/w_{H_n}$; insert $|L'(0, \chi)| = C_{\chi}(0)L'_{\text{prim}}(0, \chi)$ (all factors positive). The torsion is Lemma 6.11. $\square$

The same proof, with $\text{Cl}(\mathcal{O}_n)$ replaced by a quotient $B = \text{Cl}(\mathcal{O}_n)/A$ and f by the coset sums $\sum_{\mathfrak{c} \in bA} f(\mathfrak{c}) = \log |N_{H_n/H^A} G_{\mathfrak{c}}|$, gives the index in every subfield cut out by a subgroup:

Theorem 6.12 (layer indices). *For a subgroup $A \leq \text{Cl}(\mathcal{O}_n)$ with fixed field H^A , $B = \text{Cl}(\mathcal{O}_n)/A$, the ratio group $\mathcal{V}^A = N_{H_n/H^A}(\mathcal{V}_n) = \langle \theta^{(b)}/\theta^{(b')} \rangle$ of the coset products $\theta^{(b)} = \prod_{\mathfrak{c} \in bA} G_{\mathfrak{c}}$ satisfies*

$$[\mathcal{O}_{H^A}^{\times} : \mu(H^A)\mathcal{V}^A] = \frac{24^{|B|-1} \prod_{\psi \in \widehat{B} \setminus 1} |L'(0, \psi)|}{R_{H^A}} = \frac{4 \cdot 24^{|B|-1}}{w_{H^A}} h_{H^A} \prod_{\psi \in \widehat{B} \setminus 1} C_{\psi}(0).$$

For $A = \ker \chi_3$ the field $H^A = F = K(\theta_u)$ is the sextic of Section 6.2; it is totally complex but *not* CM (L_3 has a complex place), of unit rank 2, and Theorem 6.12 gives $[\mathcal{O}_F^{\times} : \mu_4 \mathcal{V}^A] = 576 h_F C_n(0)^2$ — the sextic companion of Theorem 6.6, whose $8 = 24/3$ is the real-cubic shadow of $24^{|B|-1} = 24^2$. For $A = \ker \chi_2$, the real statement is that $\theta_u^{(2)} := \prod_{\ker \chi_2} G_{\mathfrak{c}}/|M(n)|^{1/2}$

is a unit of $\mathbb{Q}(\sqrt{d_2})$ with $\log|\theta_u^{(2)}| = \frac{1}{2} \sum \chi_2 \log|G| = -6m_\chi \log \varepsilon_{d_2}$, i.e. $[\mathbb{Z}[\varepsilon_{d_2}]^\times : \langle -1, \theta_u^{(2)} \rangle] = 6m_\chi$ in the notation of Proposition 6.1.

The hyperbolic side has the same theorem, with a twist. The units R_f of Theorem 4.2 satisfy $R_{\tau f} = R_f^{-1}$, so in the ring class field H of discriminant $D = 1 - n^2$ over $K' = \mathbb{Q}(\sqrt{D})$ they lie in the odd units $E^- = \{u : \tau(u)u \in \mu\}$ for the involution $\tau = \sigma_\tau$, and their character sums are $-24L'(0, \chi)$ on the odd characters and 0 on the even ones (Theorem 2.10). Let $H^+ = H^\tau$ (totally complex, of degree h), $E = \mathcal{O}_H^\times / \mu$, E^+ the image of $\mathcal{O}_{H^+}^\times$, and $Q^- = [E : E^+ E^-]$.

Theorem 6.13 (hyperbolic odd index). *For every odd $n \geq 3$,*

$$[E^- : \langle R_f : f \in \text{Cl}(D) \rangle] = 24^{h/2} \cdot \frac{2^{h/2-1}}{Q^-} \cdot \frac{h_H w_{H^+}}{h_{H^+} w_H} \cdot \prod_{\chi \text{ odd}} C_\chi(0).$$

Proof. The odd functions on $\text{Cl}(D)$ are coordinatized by a transversal T of $\langle \tau \rangle$, and for $f(x) = \log|R_x|$ the twisted group matrix $(f(a^{-1}b))_{a,b \in T}$ has the odd characters as eigenvectors with eigenvalues $\frac{1}{2}S_\chi(f)$, so the odd regulator of $\langle R_f \rangle$ is $\prod_{\chi \text{ odd}} |S_\chi| = 24^{h/2} \prod_{\chi \text{ odd}} |L'(0, \chi)|$; $\zeta_H / \zeta_{H^+} = \prod_{\chi \text{ odd}} L_{\text{prim}}(s, \chi)$ gives $\prod_{\chi \text{ odd}} L'_{\text{prim}}(0, \chi) = (h_H R_H / w_H) / (h_{H^+} R_{H^+} / w_{H^+})$; and the orthogonal splitting of $E^+ E^-$ in the log-space gives $R_H / R_{H^+} = 2^{h/2-1} R^-(E^-) / Q^-$. Assemble as in Theorem 6.7. $\square$

At the nine odd levels $5 \leq n \leq 21$ the index is computed exactly with PARI (τ from `nfgaloisconj` as the unique automorphism of H/K' inverting every R_f , $E^\pm$ as kernels on the unit lattice, H^+ from a primitive element, Q^- with the action of τ on $\mu(H)$) and agrees with Theorem 6.13 in every case, all fields of degree ≤ 16 certified: 4, 12, 576, 96, 288, 663552, 768, 663552, 191102976, with $Q^- = 1, 1, 2, 2, 2, 8, 2, 4, 32$ — equal to $2^{h/2-1}$ except at $n = 19$, so Q^- is a genuine invariant — and $\prod C = 16$ at $n = 15$, 4 at $n = 17$ (multipliers through exact projections to the smaller orders). At $n = 21$ ($D = -440$, $\text{Cl} \cong \mathbb{Z}/6 \times \mathbb{Z}/2$, $h_H = h_{H^+} = 4$) the index is 24^6 exactly, and in the sextic layer cut out by an odd character of order 6 the coset units $N_{H/F}(R_f)$ have index $27648 = 2 \cdot 24^3 = 24^3 h_F / h_{F^+}$ in $E^-(F)$ ($h_F = 8$, $h_{F^+} = 4$, $Q_F^- = 4$; unconditional) — the hyperbolic index datum asked for in Remark 6.17.

Proposition 6.14 (the record; computational). *With PARI/GP 2.15.4 [28] (`bnfinit` at 120 digits, `bnfcertify` successful at every level $n \leq 15$, so those rows are unconditional; $n = 23$ is GRH-conditional), the independent evaluation of Lemma 2.3 for the $L'(0, \chi)$, and the certified integer polynomials D_n of Theorem 2.6 factored inside H_n by `nfroots`: the integer $24^{h-1} \prod |L'(0, \chi)| / R_{H_n}$ is certified with ≥ 105 spare digits, equals the right side of Theorem 6.7, and equals the determinant of the exact exponent matrix of the units G_c / G_1 in PARI's fundamental-unit basis (`bnfisunit`):*

n	$\text{Cl}(\mathcal{O}_n)$	h	w	h_{H_n}	$\text{Cl}(H_n)$	$\prod C_\chi(0)$	index	$\text{Smith invariants of } E_n$
3	$\mathbb{Z}/2$	2	12	1	1	1	2^3	(8)
5	$\mathbb{Z}/2$	2	4	1	1	1	$2^3 3$	(24)
7	$\mathbb{Z}/4$	4	4	1	1	1	$2^9 3^3$	(24, 24, 24)
9	$\mathbb{Z}/6$	6	12	1	1	4	$2^{17} 3^4$	(96, 24, 24, 24, 8)
11	$\mathbb{Z}/6$	6	4	1	1	1	$2^{15} 3^5$	(24, 24, 24, 24, 24)
13	$\mathbb{Z}/6$	6	4	3	$\mathbb{Z}/3$	1	$2^{15} 3^6$	(72, 24, 24, 24, 24)
15	$\mathbb{Z}/4 \times \mathbb{Z}/2$	8	12	2	$\mathbb{Z}/2$	32	$2^{27} 3^6$	(384, 48, 48, 24, 24, 24, 8)
23	$\mathbb{Z}/12$	12	4	12	$\mathbb{Z}/6 \times \mathbb{Z}/2$	1	$2^{35} 3^{12}$	(144, 48, $24^{\times 9}$)

The same run re-derives the eight-level cubic table of Theorem 6.6 at its primitive levels by **bnfunit** in $L_3 = \mathbb{Q}(\theta_u)$ ($\theta_u = \pm\eta^8, \eta^8, \eta^{24}, \eta^{16}$ at $n = 9, 11, 13, 23$, η PARI's fundamental unit), and the sextic indices 576, 576, 1728, 2304 ($h_F = 1, 1, 3, 4$).

Remark 6.15 (the 24-th-root saturation). The Smith invariants say more than the index. Let $\mathcal{W}_n = \{u \in \mathcal{O}_{H_n}^\times : u^{24} \in \mu(H_n)\mathcal{V}_n\}$. At the levels with $w = 4$ every G_c/G_1 is a 24-th power in $\mathcal{O}_{H_n}^\times$ up to a root of unity (an exact 24-th power at $n = 13, 23$), at the levels with $w = 12$ an 8-th power; and at every computed level $[\mathcal{O}_{H_n}^\times : \mathcal{W}_n] = \prod_i d_i / \gcd(d_i, 24)$ equals $h_{H_n} \prod_\chi C_\chi(0) - 1, 1, 1, 4, 1, 3, 64, 12$ — with $\mathcal{O}_{H_n}^\times/\mathcal{W}_n \cong \text{Cl}(H_n)$ as abstract groups at the primitive levels $n = 13$ ($\mathbb{Z}/3$) and $n = 23$ ($\mathbb{Z}/6 \times \mathbb{Z}/2$).

Conjecture 6.16 (saturated Kubert–Lang form). *For every $n \geq 2$, $[\mathcal{O}_{H_n}^\times : \mathcal{W}_n] = h_{H_n} \prod_{\chi \neq 1} C_\chi(0)$; at levels all of whose characters are primitive, $\mathcal{O}_{H_n}^\times/\mathcal{W}_n \cong \text{Cl}(H_n)$ as $\mathbb{Z}[\text{Gal}(H_n/K)]$ -modules.*

Given Theorem 6.7, the first statement is equivalent to $[\mathcal{W}_n : \mu\mathcal{V}_n] = 4 \cdot 24^{h-1}/w_{H_n}$, a statement about which 24-th roots of the Δ -quotients — Siegel units — lie in H_n [10, Ch. 11–12]; the second is of Gras type. The comparison of $\mathcal{V}_n$ with the Stark unit of H_n [26, 7], whose group has index h_{H_n} in $\mathcal{O}_{H_n}^\times$ modulo torsion at prime conductor, is now a precise question: Theorem 6.7 puts $\mathcal{V}_n$ at index $4 \cdot 24^{h-1}/w$ below it, and Remark 6.15 suggests that the 24-th roots close the gap exactly.

Remark 6.17 (the hyperbolic side). The hyperbolic R_f satisfy $R_{\mathfrak{r}f} = 1/R_f$, so their projections vanish on every character with $\chi(\mathfrak{r}) = +1$; since cubic characters are even on the 2-torsion class $\mathfrak{r}$, the hyperbolic cubic layer degenerates identically. The hyperbolic Robert index lives entirely on the odd characters: the quadratic layer of Proposition 6.1, and the sextic $\chi_2\chi_3$ -layer at levels such as $n = 21$, where the odd-part coset products of the R_f are units by Theorem 4.2; their index is Theorem 6.13 and the record following it.

7. OUTLOOK

Four continuations are genuine, in the sense that the data needed to start them is already computed and certified.

- (1) *The cocycle of the first-power units* (Open problem 5.4). Determine $\zeta(\sigma, f)$ by Shimura reciprocity for γ_2, γ_3 (Gee–Stevenhagen, Schertz N -systems [20]) applied to the hybrid object w_f ; the certified table $m(n)$ of Proposition 5.3 is a sharp target, and the η^4 -form (5.1) suggests a Dedekind-sum formula in (r_0, s_0) .
- (2) *The genus-refined Gross–Zagier exponent law*. The tables of Propositions 3.3 and 3.7 express every genus-character j -sum as $k \log \varepsilon_d + \sum_p e_p \lambda_p$ with $e_p \in \{\pm 3, \pm 6, \pm 9\}$ for Σ_0 and even for Σ_{1728} , supported exactly on $\text{GZ}(D, -3)$ resp. $\text{GZ}(D, -4)$. Gross–Zagier’s $\epsilon(\mathbf{n})$ -formula for the genus-character weighted factorization of $(j - j')$ -norms [6], in the conductor-degenerate refinement of Lauter–Viray [13], should prove the exponents and signs. This one finite-place computation would settle at once the exact denominator valuations of the phase [8][§6.7], the non-split part of the λ_n -law (Corollary 4.14 reduces it to the β -collision slopes), and the general law of the conductor Euler corrections at square conductors.
- (3) *The hyperbolic 2-adic invariant*. Theorem 6.13 reduces the hyperbolic index to h_H/h_{H^+} , the roots of unity, the multipliers and $Q^- = [E : E^+ E^-]$; the data ($Q^- = 2^{h/2-1}$ except at $n = 19$) ask for the cohomological law of Q^- (H^1 of $\langle \sigma_\tau \rangle$ on the units, i.e. ambiguous classes of H/H^+) and for the levels $n \geq 23$.
- (4) *Stark recognition for characters of order > 2* . The certified coset cubics and quartics (Remark 2.7, Section 3.4) pin $L'(0, \chi_3), L'(0, \chi_4)$ as logarithms of explicit algebraic numbers in the cubic and quartic subfields of the ring class field; identifying them with the Stark units of those fields — a theorem over an imaginary quadratic base through elliptic units [26], [20, Ch. 6–7] — would turn the certified non-fits of Proposition 3.8 into positive closed forms, and is the natural route to the Siegel-unit refinement of Conjecture 6.16.

What is new, with anchors. Each item names the repository document and section where the full record (proofs, data, script output) lives.

- The two Kronecker limit formulas for a geometric invariant of circles: Theorems 2.5 and 2.9 (`phase-kronecker-limit.md`, §2 and §5).
- The genus closed forms with the field $\mathbb{Q}(\sqrt{n})$ and the exact GZ-supported dressings: Section 3 (`phase-kronecker-limit.md`, §4 and §6).
- The unit theorem on all strata: Theorem 4.2 (`schmidt-units.md`, §2).
- The per-class valuation law and the split ladder: Theorem 4.11 and Corollary 4.13 (`schmidt-units.md`, §3).
- The first-power units, with $w = -\varepsilon_{12}^{\pm 1}$ at $n = 7$: Section 5 (`schmidt-units.md`, §4).
- The Robert index $8h_{L_3}C_n(0)$: Theorem 6.6 (`schmidt-units.md`, §5).

Open, with the same anchors: the cocycle law $m(n)$ (`schmidt-units.md`, §4); the exponent law of the dressings (`phase-kronecker-limit.md`, §8, open problems 1–2); the sextic index (`schmidt-units.md`, §5); and Stark recognition (`phase-kronecker-limit.md`, §8, open problem 3).

APPENDIX A. MACHINE VERIFICATION

Every finite claim of this paper — classwise laws, the limit-formula identities against the independent evaluation, integer polynomial coefficients, exact factorizations in quadratic fields, Newton polygons, the coherence table, fundamental units and indices, and the certified non-fits — is verified by one of three standalone Python programs in the repository accompanying the paper (`python3` with `mpmath` [9] and `sympy` [14]), each re-run from a clean checkout for this draft; the statements imported from [8] (the closed forms, $\omega \equiv 1$, the transport lemma, the Δ -mass law) are covered by the scripts catalogued in [8][App. A]. Runtimes are on a single commodity core.

The certification guard rails of [8][App. A] are enforced throughout: exact integer or rational arithmetic wherever a statement is finite (Hermite normal forms, ideal arithmetic in quadratic fields with Hensel-lifted valuations, exact factorization); a numerical value is accepted as an integer or rational only with at least $\max(20, \text{dps}/5)$ spare digits in the absolute-error sense; working precisions are set at run time, never at module import; and integer-relation searches appear only in negative statements, in the information-theoretically safe regime (a relation would be accepted only if its information content were at most a quarter of the working precision and its residual at rounding level). No claim of the paper rests on an integer-relation search.

script / phase	verifies	precision, spare digits	time
phase_klf.py --selftest	Euclidean $n = 3, 5, 7, 9, 11, 13$ and hyperbolic $n = 9, 11, 13, 15$ at 250 digits; Euclidean $n = 15$ at 400 digits		4 min 39 s
classwise laws	Lemmas 2.4, 2.8, constants included, every class	dev. $\leq 1.4 \cdot 10^{-248}$ ($1.4 \cdot 10^{-399}$ at 400d)	
limit formula	$\sum \chi \log g = -12L'(0, \chi)$ against Lemma 2.3 (600–4600 terms), every character	res. $\leq 6.2 \cdot 10^{-249}$	
master identities	Theorems 2.5, 2.9, every character; even hyperbolic sums vanish	res. $\leq 1.8 \cdot 10^{-248}$; $\leq 7.7 \cdot 10^{-250}$	
anchor	(2.3) against certified H_{-4n^2} and the closed form of $M(n)$	res. $\leq 9 \cdot 10^{-249}$	
D_n	Theorem 2.6: integer coefficients, $D_n(0) = (-1)^h M(n)$, uniform Stark law; coset cubics of Remark 2.7	spare 226–248 (358 at $n = 15$)	
genus factorizations	(3.1) as exact integer identities, $m \leq 300$; closed forms (3.2) against Lemma 2.3; hyperbolic table of Proposition 3.6; even-character factorizations	exact; res. $\leq 4.2 \cdot 10^{-250}$	
Σ -factorizations	coset products as integer quadratics; every A/B factorization re-multiplied exactly in $\mathbb{Q}(\sqrt{d})$ (Propositions 3.3, 3.7); GZ support	spare 174–238 (315–317 at 400d)	
non-fits	Proposition 3.8: safe PSLQ, height 10^6 , tolerance 10^{-200} (10^{-320} at 400d)	information $\leq$ dps/4	
schmidt_units.py --selftest	four phases		4 min 0 s
phase 1	Lemma 4.1 classwise, odd $n \leq 21$; level polynomials of Proposition 4.9 (integer, palindromic, constant term 1, regression-checked, irreducible by exact factorization); stratum polynomials and the $R \equiv 1$ collapse of Proposition 4.10	250–dev. 420d; $\leq 6.5 \cdot 10^{-419}$; spare 237–385	
phase 2	Proposition 4.12: single-slope Newton polygons of the certified D_n at every $p \mid n$, $n = 49$; $n \leq 18, 21, 25, 27, 49$; split-ladder witness at $n = 15$; inert ladders of Corollary 4.14 re-derived from certified slopes	250d (460d at 140–249)	
phase 3	Proposition 5.2(1)–(3) at every class; the table $m(n)$ of Proposition 5.3, each entry re-certified; coherent-level polynomials	250–500d	
phase 4	Proposition 6.1 projections (res. $\leq 6.2 \cdot 10^{-249}$); at all eight cubic levels: unit-normalized coset cubics, root descent $k \leq 40$ with the Friedman cap, $L'(0, \chi_3)/R_{L_3}$ as certified integers, $\log \theta_u = -8L'$, the index, and the regression record of Theorem 6.6	250d; spare 14–43 s 249–250 per level	
robert_index_full.py --selftest	Euclidean $n = 3, 5, 7, 9, 11, 13, 15$ (--with-23: $+n = 23$), hyperbolic $n = 5, \dots, 21$; needs PARI/GP 2.15.4	150d	78 s (+55 s)
determinant	Lemma 6.8 on the h numbers $\log G_c $, rel. signed, vs $2^{h-1} \prod S_\chi$; $S_\chi = -12L'(0, \chi)$ vs Lemma 2.3	spare ≥ 148	
multipliers	primitive levels (exact kernels), $C_\chi(0)$ by Lemma 6.10, checked against $L'(0, \chi)/L'(0, \chi_{m_\chi})$	spare ≥ 148	

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